ACE MANUFACTURING SYSTEMS LTD

INDEX - MACHINE HISTORY

AMS — In Pursuit of Excellence

MACHINE NUMBER : μ26-848

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	MACHINE DISPATCH CLEARANCE	F-QA-FA -16	✓
2	PACKING LIST / DISPATCH DETAILS	F-TSG-18 & F-PAC-01	✓
3	MINUTES OF MEETING	F-TSG-17	NA
4	WORK ORDER	F-TSG-15	✓
5	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
6	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
7	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
8	CHECK LIST - MECHANICAL	F-QA-FA-07	✓
9	MACHINE HISTORY - MECHANICAL	F-QA-FA-10	✓
10	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
11	MACHINE HISTORY - ELECTRICAL	F-QA-FA-14	✓
12	FINAL ASSEMBLY REPORT - MECHANICAL	F-QA-FA-05/15	✓
13	GEOMETRICAL TEST REPORT	DS-DSN-PROD-TC 01~ 06	✓
14	PRACTICAL TEST REPORT	F-TSG -34	✓
15	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
16	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
17	HYDRAULIC ASSEMBLY REPORT (OPTIONAL)	F-QA-IPR-HYD-01	✓
18	BALL BAR REPORT	F-QA-FA-02	✓
19	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
20	GEAR BOX / ATC ASSEMBLY	F-QA-IPR-GBA- 01 / SUPPLIER REPORT	✓
21	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	Not filed
22	BASE BUILDING	F-QA-IPR-BBA- 01/ 02	Not filed.

Date : 18/12/12 CHECKED BY APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

MACHINE DISPATCH CLEARANCE

AMS
In Pursuit of Excellence

M/c No.: M26-848	CUSTOMER : M/s SUASTIKA AUTOMATICS.
MODEL: WINNER	DATE : 18/12/12

DEVIATION DETAILS

NIL

ASSEMBLY IN CHARGE | QUALITY IN CHARGE | AUTHORIZED SIGNATURE

CORRECTIVE ACTION (BY CSG)

RESPONSIBLE PERSON : SIGNATURE OF CSG HOD

Note : Customer Support Group (CSG) will confirm the closure of deviation and Return to QA.

Wimmer
A/G

ACE MANUFACTURING SYSTEMS LTD.

MACHINE DISPATCH CHECK POINTS

AMS — In Pursuit of Excellence

CUSTOMER'S NAME: Swastika Automotics, Faridabad.

M/C No.: M26-848 Wimmer APC

DATE: 20-12-012 5:30PM

S.No	DESCRIPTION	YES/ NO/NA	S.No.	DESCRIPTION	YES/ NO/NA
1	Packing list available	Yes	21	Chip Trolley cleaned	NA
2	Hydraulic / Pneumatic Counter Balance Cylinder removed	NA	22	Standard Spares & m/c manual available	Yes
3	Z- Motor Cover Removed & Packed	NO	23	Coolant tank assly. packed	Yes
4	Side door handles removed & Packed	NO	24	Index Table Cover packed properly	NA
5	Base guard removed & Packed	NO	25	Oil skimmer bin cleaned	NA
6	Coolant Gun tied properly	Yes	26	Voltage Stabilizer packed	NA
7	Air gun tied properly	NA	27	Machine Platform cleaned	NA
8	Leveling Screws, Nuts removed & Packed	Yes	28	Wanner Pump & cover cleaned & put back	NA
9	Leveling blocks (100)/180/220/300 Ht. Qty: - 4/6/7/(8)/11/20	Yes	29	Customer components packed properly	NA
10	Automatic Tool Changer	Yes	30	Index Table packed properly	NA
11	APC Assembly packed Properly	NO	31	Chiller unit cleaned	NA
12	Pallet removed & Packed	NO	32	Rotary Table packed properly	NA
13	APC leveling screws - 1set	Yes	33	Toolings packed properly	NA
14	Power pack stand	yes	34	Tail Stock packed properly	NA
15	Lub Tank oil drained - retain 2 ltrs.	Yes	35	Practical Test piece packed	NA
16	Machine oil Bin cleaned	Yes	36	Fixture/s packed properly	NA
17	Coolant tank mesh cleaned & put back	Yes	37	Air removed from cover with Vaccum cleaner	NA
18	Chip Bin cleaned & put back	Yes	38	Desiccant Bags Hung	NA
19	Chip Bin mesh cleaned & put back	Yes	39	Fumigation Certificate available	NA
20	TPM points stickers fixed properly	NA	40		

REMARKS / SHORTAGES IF ANY:

Nil

ASSEMBLY	STORES	FIXTURE	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY
				Tayama	

ACE MANUFACTURING SYSTEMS LTD.

AMS
In Pursuit of Excellence

DISPATCH CHECK POINTS

CUSTOMER'S NAME : Swastika Automotics,

MACHINE NUMBER : H26-848 **DATE :** 20-12-012

S.No	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	Yes	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	Yes	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE (in side the cabinet)	— yes —	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON M/c	Yes	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	— NO —	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	CHECK ITEMS AS PER PACKING LIST	Yes	
22	WOODEN PACKING PROVIDED BELOW THE APC STATION.	— Yes —	
23	WOODEN support given below the grundfoss pump Mounting bracket	———	this proposal is w.e.f 23/4/2011
24.	grundfoss pump Removed and packed seperately	Yes/NA/No	
25	MACHINE LIFTING PROCEDURE SKTCH PASTED ON THE GUARD	— Yes —	

Jayanna. V. Gopal. KM

ATTENDED BY | QUALITY ASSURANCE | DISPATCH - INCHARGE



CONTROLLED COPY

Sign: M Date: 1/8

ACE MANUFACTURING SYSTEMS LTD

PACKING CASE IDENTIFICATION LIST

MACHINE NUMBER: M26-848 DATE: 30-12-012

SL.NO	CASE-1 CONTENTS	QUANTITY
1	M/c assembly	01NO
2	Std spares and M/c Manuals	01Box
3	Spin-12 oil	01Liter
4	Leveling screws and Lock nut	06+6NOS
5	Leveling blocks-100MM	08NOS
6	Apc Leveling pad	01Setts
7	Power pack	01NO
8	coolant gun	01NO
9		
10		
11		
12		

CASE-2 CONTENTS

1	Coolant tank assembly	01NO
2	Coolant Pump	01NO
3	Mesh	08NO
4	Chip bin	01NO
5	Chip bin Mesh	02NOS
6	Chucker plate	01NO
7	oil bin	01NO
8	Power pack Stand	01NO
9		
10		
11		
12		

CASE-3 CONTENTS

1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

CHECKED BY : Fayaziva

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST ANNEXTURE-II

Doc No	PAN#000199	Date	18/12/2012	M/c No:	M26-848
M/s	SWASTIKA AUTOMATICS.,FARIDABAD				

Sl. No	Description	Qty	Remarks	PART NO
1	O'Ring ø8.1 x 1.6	6		S15-005-00
2	O'Ring ø24.5 x 3.0	6		S15-005-17
3	O'Ring ø57.6 x 2.4	6		S15-005-18
4	O'Ring ø19.1 x 1.6	2		S15-003-33
5	O'Ring ø11.1 x 1.6	4		S15-005-01
6	O'Ring ø51.6 x 2.4	2		S15-003-70
7	Cord of O'Ring ø3 X 2 Mtrs	1		S15-015-20
8	Spanner Double Ended 10 X 11	1		S31-001-01
9	Spanner Double Ended 12 X 13	1		S31-001-02
10	Spanner Double Ended 18 X 19	1		S31-001-03
11	Spanner Double Ended 30 X 32	1		S31-001-04
12	Spanner Double Ended 36 X 41	1		S31-001-11
13	Allen Key Set	1		S31-002-101
14	Extended Allen Key (A/F 8)	1		C99-5106-8904-00
15	Steel Ball Ø8	8		S09-015-02
16	Fuse Set (Fanuc Make)	1		S22-010-11
17	Glass Fuse Set (Kept in Electrical Cabinet)	1		*********
18	Lubrication Oil (Spin-12 / Process Oil-12)	1		S04-005-07-CAN
19	Pull Stud BT-40 (Mas-403)	12		C74-3102-8114-01
20	FANUC Series Oi Model D / Oi Mate-Model D USER`S MANUAL	1		D00-003-00-27
21	FANUC Series Oi Model D / Oi Mate-Model D MAINTENANCE MANUAL	1		D00-003-00-28
22	Fanuc AC SERVO & SPINDLE MOTOR Bis series /alpha series MAINTENANCE MANUAL	1		D00-003-00-03
23	Operating Instructions And Maintainance Manual	1		D00-003-00-05
24	Spare Parts Manual	1		D00-003-00-04
25	Air Cooler Manual	1		D00-003-00-06
26	Electrical Documentation Manual	1		D00-003-00-00-000-E0
27	Machine Back Up CD (SOFT COPY)	1		D00-003-00-00-000-E3
28	Maintenance Service Note (Fanuc Warranty Card)	1		D00-003-00-23

Packed by:..... Authorized Sign.....

Printed Date & Time: 18/12/2012 5:12:17 PM

ACE MANUFACTURING SYSTEMS LIMITED.	AMS In Pursuit of Excellence
DISPATCH ADVICE	

Sl. No.	Description Of Dispatch details	D.No : DNo#009439
1	Customer Name	SWASTIKA AUTOMATICS, [Party Id – SWAAU001]
2	Telephone No.	0129 – 4052432 / 4053432 / 9810277729
3	Fax No.	0129 – 4052432
4	E-Mail	swastikaautomatics@touchtelindia.net
5	Customers ECC No.	
6	Customers Sales Tax No. & Date	CST: TIN: 06841304337
7	Sales Order no.	MO10002908
8	Machine Model	WINNER
9	Machine No.	M26-848
10	Invoice Address	SWASTIKA AUTOMATICS, [SWAAU001] 2990, BLOCK-A, GALI NO.1, S.G.M. NAGAR, NH-3, N.I.T, FARIDABAD – 121 001
11	Place of Delivery	Default Address
12	Purchase Order No. & Date	SW/AMS/06 Date:11/12/2012
13	P.O Value (Excluding Taxes & Duties)	Rs.23,10,000.00
14	T.ansportation	Door delivery through any reliable carriers and freight to-pay basis.
15	Insurance	Will be arranged by the customer. Dispatch details to be faxed to the customer.
16	Scheduled Dispatch	20/12/2012
17	Payment	Refer payment collection detail sheet
18	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch Shortage list to be made while preparing machine invoice All Pending Points to be attended before machine dispatch

NOTE: ENCLOSED ROAD PERMIT

Copy: Production/Electrical/Design/Purchase/Accounts/Dispatch/TSG

F-TSG-19-R1

For ACE MANUFACTURING SYSTEMS LTD.

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST

PACKING LIST NO.	DNo#009439	Date: 20/12/2012
SALES ORDER NO.	MO10002908	Date: 12/12/2012
CUSTOMER	SWASTIKA AUTOMATICS, FARIDABAD	
P.O. NO.	SW/AMS/06 Date : 11/12/2012	

Sl. No.	Item Id	Model	Qty
1	M26-848	WINNER	1.000

Sl. No.	Description	Short Supply?
1	CNC Vertical Machining Center Model "WINNER"	
	Spindle nose taper 7/24 No.40	
	Spindle power (Fanuc) 15 Min./Cont. Rating - 5.5 / 3.7 KW	
	AC Servo axes drive (Fanuc)	
	12 Station tool magazine	
	Tool shank BT-40	
	Hardened Guideways for all the three Axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ballscrew stretching for X, Y & Z Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Standard CNC features of Fanuc 0IMATE-MD with Colour LCD	
	Spindle speed of 60 - 6000 rpm (Indirect drive)	
	Rapid traverse of 24 / 24 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Spindle with MAS - 403 type pull stud clamping system	
3	Flood coolant system around the spindle	
4	Chip flushing system on the base	
5	Coolant gun	
6	Automatic Pallet Changer (Pallet Size: 600 x 385) - Linear type to RH side	
7	MAS - 403 Pull studs (12 nos.)	
8	SERVO SPIN - 12 lubricating oil (1 liter)	
9	1 set of glass fuses(inside the cabinet)	
10	Electrical documentation manual & back up CD	
11	Geometrical test report	in Side the Cabinet
12	NAS test report	
13	Laser Calibration test report	
14	Heavy cut & Rigid tapping test report	

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For ACE MANUFACTURING SYSTEMS LTD.

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: WINNER

Sl. No.	Description	Unit	WINNER (WITH APC)
1	**AUTOMATIC PALLET CHANGER**		
1.1	PALLET SIZE	mm	600 x 385
1.2	T - SLOT DIM.	N x W x P	3 x 14 x 100
1.3	NO. OF PALLETS	Nos.	2
1.4	WEIGHT ON PALLET MAX	Kgf.	150
1.5	PALLET CHANGE TIME	Sec.	18
1.6	PALLET HEIGHT FROM FLOOR	mm	1100
2	**MACHINE CAPACITY**		
2.1	X - AXIS TRAVEL	mm	525
2.2	Y - AXIS TRAVEL	mm	400
2.3	Z - AXIS TRAVEL	mm	350
2.4	DIST. FROM SPINDLE FACE TO PALLET TOP	mm	170 - 520
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	470
2.6	COOLANT SUPPLY - FLOOD COOLANT AROUND THE SPINDLE	lpm	50
3	**SPINDLE**		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - INDIRECT DRIVE	rpm	60 - 6000
3.3	SPINDLE BEARING DIAMETER	mm	60
4	**FEED**		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	24 / 24 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	GUIDEWAYS FOR X, Y & Z AXES		Hardened
5	**AUTOMATIC TOOL CHANGER**		
5.1	NUMBER OF TOOLS	Nos.	12
5.2	MAX. TOOL DIA	mm	125
5.3	TOOL SELECTION		Bi - Directional
5.4	TOOL LENGTH MAX.	mm	250
5.5	TOOL WEIGHT MAX.	Kgf	6
5.6	CHIP TO CHIP TIME ADJACENT TOOL	Sec.	5.5
5.7	TOOL CHANGE SYSTEM		DISC ARMLESS
5.8	TOOL SHANK TYPE		BT-40
5.9	PULL STUD		MAS - 403
6	**ACCURACY WITH ROTARY ENCODER**		
6.1	POSITIONING FOR X, Y & Z AXES AS PER VDI DGQ 3441	mm	0.015
6.2	POSITIONING FOR X, Y & Z AXES AS PER ISO 230-2	mm	0.010
6.3	REPEATABILITY FOR X,Y&Z AXES AS PER VDI DGQ 3441	mm	± 0.005
6.4	REPEATABILITY FOR X, Y & Z AXES AS PER ISO 230-2	mm	± 0.003
7	**MOTORS**		
7.1	SPINDLE POWER 15 MIN./CONT. (FANUC)	kW	5.5 / 3.7
7.2	FEED MOTOR X, Y & Z AXES (FANUC)	Nm	7 / 7 / 20

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: WINNER

Sl. No.	Description	Unit	WINNER (WITH APC)
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	2500
8.2	MACHINE DEPTH	mm	2900
8.3	MACHINE HEIGHT	mm	2875
8.4	MACHINE WEIGHT - NET	Kgf	4360
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	12 / 14
8.6	CNC CONTROL (WITH 8.4" COLOUR LCD)		FANUC 0IMATE-MD
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
9.3	HYDRAULIC POWER PACK	litr	32
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALLSCREW STRETCHING FOR X,Y & Z AXES		
10.13	TOOL LIFE MANAGEMENT		
10.14	AIR DRIER UNIT (SMALL)		
11	**STANDARD CNC FEATURES OF FANUC-0IMATE-MD**		
11.1	THREE CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 3 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF THREE AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

AMS — In Pursuit of Excellence

MODEL: WINNER

Sl. No.	Description	Unit	WINNER (WITH APC)
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y CO-ORDINATES)		
11.15	TOOL OFFSET MEMORY - 400 NOS.		
11.16	TOOL LENGTH COMPENSATION		
11.17	PART PROGRAM MEMORY 1280 MTS (512K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
11.36	SPINDLE LOAD DISPLAY		
12	**MACHINE FEATURES / ACCESSORIES - ADDITIONALLY ORDERED BY M/S SWASTIKA AUTOMATICS, FARIDABAD.**		
12.1	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
12.2	CHIP FLUSHING SYSTEM ON THE BASE		
12.3	COOLANT GUN		
12.4	AUTOMATIC PALLET CHANGER (PALLET SIZE 600 x 385)		

ACE MANUFACTURING SYSTEMS LTD

MACHINE SEQUENCE RUNNING TEST REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : μ126-848

Preconditions for Machine before starting Machine Sequence Running

1. Machine should be fitted with all accessories as per Work Order ✓
2. Machine should be cleared from all Electrical & Mechanical Remarks ✓
3. Machine should be run with Sufficient Coolant. ✓
4. ATC should be full with tool holders ✓
5. No Bypass of any Safety switches and cabinet sealed. ✓

Date	Time	No of Hours Running	Date	Time	No of Hours Running	Remarks / Alarm Observed
18.12.12	2 A.no PM	1 ✓			17 ✓	
		2 ✓			18 ✓	
		3 ✓			19 ✓	
		4 ✓			20 ✓	
		5 ✓			21 ✓	
		6 ✓			22 ✓	
		7 ✓			23 ✓	
		8 ✓			24 ✓	
		9 ✓				
		10 ✓				
		11 ✓				
		12 ✓				
		13 ✓				
		14 ✓				
		15 ✓				
		16 ✓				

Nature of Remarks Related to	1.Hydraulic leakage	2. Coolant leakage	3.Electrical alarm	4. ATC	5. Spindle	6. Axes	7. Others

Note : In case M/c is stopped due to any reason, write the Sl. No. of the above remarks against corresponding number of hours, set right the M/c & continue to run for a total period of 24 hrs.

In charge - Mechanical **In charge - QA**

ACE MANUFACTURING SYSTEMS LTD

SPINDLE TEMPERATURE STABILIZATION REPORT

AMS — In Pursuit of Excellence

M/c. NUMBER : M26-848	MAXIMUM RPM : 6000	DRIVE: ID/DD/IDCTS/DDCTS
SPINDLE SL No. : 12-142	PULLEY RATIO : 1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2	

Note : Refer TM-QA-07 for test procedure. **Recommended Temp. Rise : 20°C max. above ambient**

% of current drawn at max RPM of the spindle : (Recommended max : DD : β3 = ≤ 15%, β6 / α6 & above = ≤10%. ID : β3 = ≤ 20%, β6 / α6 & above = ≤15%.	Measured = 8 %
Spindle noise at max RPM [Recommended : 80 dB(A) upto 10 kW, 85 dB(A) for > 10 kW]	Measured = 77 dB (A)

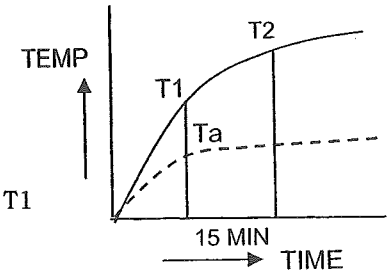
SPINDLE RPM	TIME	AMBIENT TEMPERATURE	SPINDLE TEMPERATURE
500	10:00	25°C	—
6000	11:00	26°C	34°C
	11:30	26°C	36°C
	12:00	27°C	38°C
	12:30	27°C	39°C
	1:00	27°C	40°C
	1:30	27°C	40°C
	2:00	28°C	41°C

1. The Stabilization point, Calculated as follows. Is the Temperature at which the rate of temperature rise over 15 minutes is less than 5%.

= [(T2 – T1) / (T1 –Ta)] X 100

= < 5%

T1 = Previous Temp., T2 = Temp. after 15 Min & Ta = Ambient Temp. at T1



2. RUN SPINDLE AFTER STABILIZATION FOR 3 HOURS.

10/12/12 Date	Checked By

F-QA-FA-11-R0

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

MACHINE HEALTH CHECK POINTS

MACHINE NUMBER : VI26-848 **DATE :** 10/12/12

I	**AXIS**	
1	Ensure plugs are fitted for L M Guide ways.	—
2	Ensure effective wiping of saddle cover wipers.	T-COVER
3	Ensure the lube timer is set 12 min for HG and 20 min for LM Guide ways.	—
4	Ensure leveling bolts are fixed with locknut.	✓
5	Check the Table tenon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm)	✓
6	Ensure butt reference function.	✓
7	Check all the axes reference arrows are matched.	— NA —
8	Check Emergency push button function of Machine / APC / Double Hand	✓
9	Ensure marking of all screws with Yellow paint marker	✓
10	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	— NA —
11	Ensure smooth movement of sliding door without play and unhealthy noise	✓
II	**SPINDLE**	
1	Ensure air jet through all the holes (4/8 holes) at oriented spindle position	✓
2	Ensure supply of air to spindle Air Purge system	— NA —
3	Check the tightness of push stud.	✓
4	Ensure grub screws are fitted to spindle flange.	✓
5	Check spindle orientation position and note down the value. (Parameter No. : 4031 for Fanuc controller, ... for Siemens controller)	1601
6	Ensure spindle balancing is completed	✓
7	Ensure presence of gap between draw bar bush & ejection screw / roller (visually or by feel)	✓
8	Check the "Tool not in spindle" proximity message.	✓
III	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Ensure oil level is max. in ATC gear box.	✓
3	Check the Tool Number display on panel board is same as in spindle	✓
4	Check tool change position limit switch for proper functioning & band width (Max. 1.5 mm on each side).	— NA —

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ACE MANUFACTURING SYSTEMS LTD MACHINE HEALTH CHECK POINTS		AMS In Pursuit of Excellence
IV		COOLANT
1	✓	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe
2	✓	Ensure proper working of float switch (by shifting the sensor upwards).
3		Check the function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).
4		Ensure drum wash in conveyors (Optional)
5		Check conveyor overload message & Direction (Optional)
6		Check oil in conveyor gear box. (Optional)
7		Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).
8	✓	Ensure CTS flow switch setting by opening the bag filter air vent ball valve (Optional)
9	✓	Ensure suction strainer & Check Valve in CTS tank & pr. gauge functioning (Optional)
10		Ensure CTS pin-locator hole is plugged (Optional)
11		Ensure that twin arm plunger & CTS pin locator screw do not foul (Optional)
12		Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)
13		Ensure High pressure air blast after CTS switched off (Optional)
V		APC
1	✓	Check pallet initialization in APC Machine
2	✓	Ensure supply of air jet to pallet rest pad cleaning.
3	✓	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise
VI		HYDRAULIC, PNEUMATIC & LUBRICATION
1		Ensure air pressure switch is working properly.
2	✓	Ensure proper setting of oil (1 drop / tool change) in FRL unit & Oil Level low message
3	✓	Check the FRL drain hose is fixed.
4	✓	Ensure identification name plates are fixed on power pack hoses.
5	N/A	Check the Hyd.oil level low message (by adjusting sensor position) & Pr.-gauge function
6	✓	Ensure hydraulic flow control valve grub locking in APC machine.
7	✓	Ensure Ball valve fixed to FRL Unit inlet
8	✓	Ensure no oil leakage in hydraulic valves & fittings
9	✓	Ensure lubrication pressure switch is set to 15 bar.
10	N/A	Ensure pneumatic quick change coupling fixed to 4th axis provision
VII		GENERAL
1	N/A	Check all the points of MOM are attended.
2	✓	Ensure coolant tank capacity engraved or marked in name plate
3		Check Hydraulic specification is as per circuit diagram
4	✓	Ensure axis calibration & ball bar parameter as per the report.
5	✓	Ensure Ball screw pitch in the machine and report is same.
6	✓	Ensure spindle Sl. No. and parameters are as per the report.

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MACHINE HEALTH CHECK POINTS

MACHINE No. : M26 - 848 ASSEMBLY TECHNICIAN : Sudeepa & Team

S.No	REMARKS OBSERVED	STATUS	SIGN
1	X axis front buffer fixing to be done	OK	
2	axis arrow mark matching to be done	OK	
3	AC unite mesh fixing to be done	B	
4	Coolent tank chip bin fixing to be done	OK	
5	Side cam pinning to be done	OK	
6	Side cam special washer fixing	OK	
7	APC Box Emergency not working	E	
8			

ASSEMBLY IN CHARGE

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CHECK LIST - MECHANICAL

MACHINE NUMBER : M26-848 **DATE :** 20/10/12

I Z-axis motor current during tool declamp (Turret type ATC) : Recommended Max : BT30-150%, BT40-180%)

A	With Z axis moved at 840 mm/min in Jog mode	
B	In Auto cycle	

II Z - axis drop during power off : 200

- ve' direction< 0.005 mm		'+ ve' <0.015mm direction	
System Initial Reading :	200.000	System Initial Reading. :	200.000
System Final Reading :	200.005	System Final Reading. :	200.007
System Diff. in Reading :	0.005	System Diff. in Reading. :	0.007
Dial Reading. :	0.002	Dial Reading. :	0.004

III % of Axis current drawn (Recommended as per Data sheet DS -QA -02)

A

Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm/m Feed rate
X	14%	17%	45%	15%
Y	19%	20%	27%	20%
Z Up	46%	48%	53%	50%
Z Down	23%	21%	21%	19%
B - Forward				
B - Reverse				

B % of Axis holding current (Recommended as per Data sheet DS -QA -02)

Axis	100% Rapid Rate		2000 mm/min. Feed Rate	
	Motor end	Brake end	Motor end	Brake end
X	16%	14%	6%	7%
Y	30%	27%	7%	9%
Z	44%	28%	32%	37%

C Over travel against Axis dead stopper in mm (Recommended as per Drawing)

Axis	Brake end	Motor End
X	541.800 ~~542.280~~	17.700 ~~17.220~~
Y	23.100 ~~28.043~~	416.890 ~~417.757~~
Z	377.400 ~~376.212~~	178.720 ~~179.688~~

IV SPINDLE

A

Push stud stroke length for Tool ejection (mm)	Recommended values	Auto	Manual
BT – 30/40 /50	Refer DS-DSN-PROD-01	0.800	0.800
HSK A50/63/100		— NA —	— NA —

B Tool clamping force-Kgf — Refer DS-DSN-PROD-01 — 600 Kg force

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F-QA-FA-07-R1

C	Axial play in the spindle (Ref. TM-QA-10 for procedure) (Recommended max.except BT50 : 0.005 mm , for BT 50 : 0.008 mm)		
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ +0.025 ⟶ 0.000	Difference = mm (R1~R2) 0.002
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ - 0.023 ⟶ 0.020	

V	Turret type ATC	
A	Check the spring lever compression at :	
	1. After the magazine butts against the dead stopper (mm)	5.100
	2. Z – axis position below zero position (mm)	2.000
	3. Recommended : BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm (i.e. 1-2)	3.000
B	Tool picking position for Turret Type ATC Recommended : BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm	62.000

VI	A.P.C	
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram)	4.3 BAR
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram)	22 BAR

VII	Cycle Time in seconds (Refer Work order for recommended value)	
	1. Check the tool to tool time	3.2 Sec
	2. Check the chip to chip time with Z axis only	4.3 Sec
	3. Check the chip to chip time with all axes	— NA —
	4. Check the pallet changing time	16.8 Sec

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CHECK LIST - MECHANICAL

AMS
In pursuit of excellence

M26-848

1. MACHINE LEVEL READING :

Along X-Axis

1/2	1/2	1/2
1/4	0	1/4
0	0	1/4

Along Y-Axis

1/4	1/4	0
0	1/4	1/2
1/4	0	1/2

2. TABLE TOP READINGS:

Along X-Axis

+3	+3	0
+6	+4	0
+12	+10	0

Along Y-Axis

-13	-5	-6
-10	-8	-5
0	0	0

3. TENON SLOT READING :

20	0

PALLET - II

-12	-0

PALLET - II

Along X-Axis

-10	-5	0
+2	+3	0
+6	+5	0

Along Y-Axis

0	0	0
+5	+8	+8
+10	+13	+10

5. MANDREL READINGS:

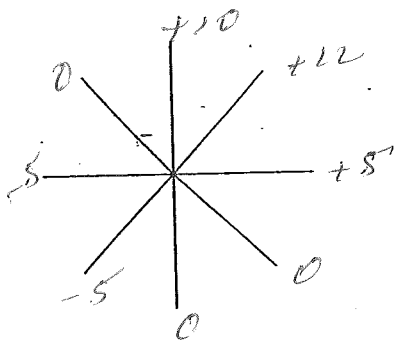
Front	Side
0	0
+4	-8

RUNOUT:

a. TOP : (0.010mm)
1. 0.003

b. BOTTOM :(0.020mm)
1. 0.015
2. 0.012
3. 0.014
4. 0.016

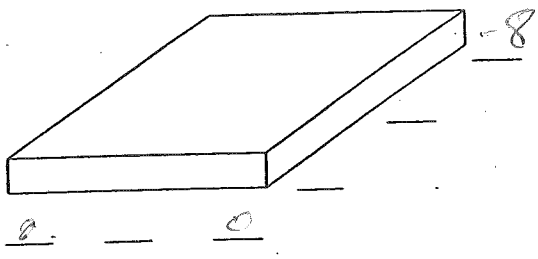
6. TURN AROUND READING:



4.COLUMN READINGS :

FRONT	SIDE
-3	+7
0	0

7. SQUARENESS READING (X & Y):



8.-PALLET HEIGHT VARIATION:

Pallet-1: 0.000

Pallet-2 : 0.010

NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01~06.

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MACHINE HISTORY - MECHANICAL

MACHINE NUMBER : M26-848 **DATE :** 10/10/12

1	ATC Details : Model	4020V / 4024V / 4030V / 4020H / 4024H /HSK A63 /HSK A100		
1.1	ATC :Turret Type/Twin Arm	Sl.No	9B12-146	
1.2	Turret type magazine Encoder	Make	Model	Sl. No
		Pragati	ATC12 / ATC16 / ATC20	
1.3	ATC Torque Motor	TM 50 / 63 / 80 / 100 / 125 / Servo ATC		
2	**BELT / COUPLING DETAILS :**			
		GT Belt (For Indirect Drive)	COUPLING (For Direct Drive)	
2.1	Make	Unitta / Power Grip	Mayr	
2.2	Model			
3	**COOLANT SYSTEM DETAILS :**			
	Make : Rajmane Industries		Model	Type
3.1	Flood Coolant Pump - Crown/Flexinox/Disc Nozzle		RKM 206 / RV 160 - 270	Centrifugal Pump
3.2	Chip Pushing Coolant Pump		RKM 206 / RV 160 - 270	
3.3	Base Flushing Coolant Pump		RKM 206 / RV 160 - 270	
3.4	Coolant Gun & Drum Wash Pump		RKM 206 / RV 160 - 270	
3.5	Oil Skimmer - Make / Model			
3.6	High Pressure Coolant Pump		Make	GF/ WP / BP
			Model	
3.7	Pressure Relief Valve		Make	Polyhydron
			Model	DPRH-10T-100-04
3.8	Rotary Joint		Make	Deublin / RIX
			Model	
4	**FLOAT SWITCH :**			
	Make		Filpro Electronics	
5	**CHIP CONVEYOR DETAILS :**			
5.1	Type of Conveyor	Scraper / Slat cum Scraper / Magnetic along X or Y axis.		
5.2	Model			
5.3	Make	Universal Conveyor		
5.4	Sl.No			
6	**CONVEYOR GEAR BOX DETAILS :**			
6.1	Model			
6.2	Make	Universal Conveyor		
7	**FILTER SPECIFICATIONS :**			
7.1	Model / Working pressure	8 /15, 50 bar / 100 bar		
7.2	Make	Universal Conveyor		
7.3	Sl.No			
8	**COIL / SCREW CONVEYOR DETAILS :**	Qty : 1 / 2		
8.1	Make	Universal Conveyor		
8.2	Model			

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MACHINE HISTORY - MECHANICAL

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9	HYDRAULIC SYSTEM :			
9.1	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi /Hymech	
9.2	Product No		QC. No : BCT 48	
9.3	Tank Capacity (Litres)		32/40 / 63 / 70	
9.4	Maximum Working Pressure		50 BAR	
9.5	HYDRAULIC PUMP	Make Rexroth / yuken /Corotech / vickers	Model RV20-SF-17-C-10	
9.6	MOTOR	Make Rotomotive / Siemens	Model 90L-4-1400 / 90S-4-1390 ,	
10	PRESSURE SWITCHES FOR HYDRAULIC SYSTEM :			
10.1	Make		Polyhydron	
10.2	Model		1PS 30-50 / 30-100 / 10 - 30 /	
10.3	Quantity		2 Nos	
11	HYDRAULIC ACCUMULATOR :			
11.1	Make	FOX / BOSCH	FOX / BOSCH	
11.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
11.3	Gas Pressure 50	Tool de clamp :	Counter balance:	
11.4	Volume (Litres)	Tool de clamp : 1.4	Counter balance : 4 or 10	
12	HYDRAULIC CYLINDERS :			
	Make	Model	Function	Qty
12.1	Island Hydraulics	Ø 40 X 22 X [125/152/180/1150]	Pallet Forward / Reverse	2 no
13	LUBRICATION UNIT DETAILS	Make	Pressure Setting	
		CENLUB / DROPCO	15 Kg/ cm2	
14	FRL UNIT			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan	MINI / MIDI	0-16 Kg/cm2	Servo Way Spin 12
15	DRYER	Make		Specification
		Trident		VXD - 2
16	PNEUMATIC VALVES			
	Application	Specs	Make	Qty
16.1	Spindle Air Jet	MFH 3-1/8	Festo	01
16.2	Spindle Air Purge	MFH 3-1/8	Festo	
16.3	ATC Pocket up & Down	JMFH 5- 1/8	Festo	
16.4	Manual Tool De Clamp	MFH 5-1/8/JMFH 5-1/2	Festo	
16.5	APC Auto Door	JMFH 5-1/8	Festo	01
16.6	CTS Air Jet	MFH 3-1/8	Festo	
16.7	Front Auto Door	JMFH 5-1/8	Festo	
		MFH 3-1/8	Festo	
16.8	Pallet Rest Pad Cleaning	MFH 3-1/8	Festo	
16.9	Fixture		Festo	

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MACHINE HISTORY - MECHANICAL

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17	PRESSURE SWITCH FOR PNEUMATIC SYSTEM			
17.1	Operating Range		2 to 15 Kg /cm2	
17.2	Make		SUCO	
17.3	Quantity		1 +	
18	Tool de clamp system:		Mechanical / Pneumatic / Hydraulic / Hydro-Pneumatic	
	Function	Make	Model	Qty
18.1	Manual Tool De Clamp - Pneumatic	Festo	DNU 80-80-PPV DNU 100-80-PPV	1
		Rotex	SCN D -100 / 60 SCN D 80 / 80	1
18.2	Manual Tool De Clamp - Hydraulic	Island Hydraulic / AMS	ϕ40 X 20 X 65	1
18.3	Manual Tool De Clamp - Hydro pneumatic	Preac	PC33-8045-AM02	1
19	PNEUMATIC CYLINDERS USED			
	Function	Make	Model	Qty
19.1	Front Auto Door	Festo		
19.2	APC Auto Door	Festo		
20	TURRET / INDEXING TABLE (OPTIONAL)			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS			8 /12 /
21	ROTARY TABLE	Make	Model	Sl.No
		NIKKEN / UCAM		
22	TAIL STOCK	Make	Model	Sl.No
23	PNEUMATIC COUNTER BALANCING			
23.1	Make	SMC / Rotex / Festo		
23.2	Cylinder Diameter :	Ø 80 / 100 / 125		
23.3	Air Reservoir used	Yes / No		
24	HYDRAULIC COUNTER BALANCING			
24.1	Make			
24.2	Model	Ø 50 X 32 X 670		
25	ADDITIONAL INFORMATION / REMARKS			
1				
2				
3				

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CABINET ENERGIZING REPORT

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CHECKPOINTS

SL No.	Description	status
1	Cabinet colour - As per work order	Yes
2	Panel layout - As per layout diagram	Yes
3	Ferruling- As per circuit diagram	Yes
4	Identification stickers /name plates/ tags - As per circuit diagram	Yes
5	Lug Crimping - Physical check	Yes
6	Terminal tightness - Physical check for loose connections	Yes
7	Ratings of main transformer& related contactor,mcb - As per circuit diagram	Yes
8	Ratings of other mcb's like TFR4 control & 230v 1ph.- As per circuit diagram	Yes
9	Ratings of mpcb's& contactors - As per circuit diagram	Yes
10	Cabinet door setting- check for lsw setting, Beading, locking,allignment	Yes
11	Health check of elements- For physical damage, enrgization, mounting/ fixing	Yes
12	Cabinet mounting - Check for Screw tightness, rubber beading if required	Yes

ADDITIONAL CONTENTS WITH CABINET

1)MACHINE HISTORY	☐	2)PRE WORK ORDER	☐
3)M/C No. PLATE	☐	4)PVC CHANNEL COVER	☐
5)PENDENT BEADINGS	☐	6)SPLIT PLATE	☐
7)SPINDLE POWER,FAN,F/B CABLE	☐	8)	☐
	☐		☐

SHORTAGE/ PENDING WORK DECLARATION

SL No.	DESCRIPTION	ACTION PLAN
1		
2		
3		
4		
5		
6		
7		

CHECKED BY:- 12/12

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

CHECK LIST - ELECTRICAL

M/C NUMBER:- M 26 - 8448 DATE:- 08/10/2012

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the work order specification for the following	✓	
	a)Servo motor	✓	
	b)Axis stroke	✓	
	c)Rapid rate & Feed rate	✓	
2	Check lub inch & Lubrication timers as required for the machine	✓	
	LM RAIL: **20Min.** HARD.GUIDWAY: **12 Min.** Grease LUB: **6 Hours.**	✓	
3	Check the connection of cable both at motor & drive end.	✓	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cable both at motor & drive end.	✓	
2	Specifications of spindle motor /speed /power is as per work order.	✓	
3	Check the spindle motor fan rotation.	✓	
4	Check the following spindle parameters	✓	
	a)Speed locking (P3735,3736)	✓	
	b)Power locking (P4028,4029)	✓	
	c)Rigid tapping (P4016#4,4044,4052,4085,5241,5261)	✓	
5	Check the spindle for following functions	✓	
	a)Maximum & Minimum speed	✓	
	b)Manual Increment & decrement	✓	
	c)Manual stop, clockwise & counter clockwise.	✓	
	d)Star/Delta switching of Contactors for required speed	NOA	
	(switching speed D340 =)	NOA	
	e)Spindle coasting	NOA	
	f)Spindle orientation @ maximmum speed.	✓	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, drive, etc.)	✓	
2	Check ATC for the following functions	✓	
	a)Magzine motor direction	✓	
	b)Arm motor direction	NOA	
	c)Tool display	✓	
	d)Tool declamp in manual mode	✓	
	e)Auto tool change	✓	
	f)Power failure conditions	✓	
3	Check for Test mode,Handheld MPG operation if **servo drive** is used.	✓	
4	Check for Test mode,parameter locking if **VFD** is used.		

IV. APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, etc.)	✓	
2	Check for pallet selection program(**O100**).	✓	
4	Check APC for the following functions	✓	
	a)MOP operations	✓	
	b)Pallet set up sequence	✓	
	c)Auto pallet change(M80) & related interlocks.	✓	
	d)Power failure conditions	✓	
	e)Fixture interlock	NOA	

CHECKED BY:- VERIFIED BY:-

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CHECK LIST - ELECTRICAL

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V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following elements		
	a)Solenoids		
	b)Pressure switches		
	c)FRL unit & auto drain		
2	Check for solenoid nut,screws & prsw caps.		

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for Hydraulic Motor / cooling fan direction.		
2	Check for cables/ hose / plugs termination & routing.		
2	Check for solenoid/pressure sw mounting screws & gasket.		
3	Check the functions of following elements		
	a)Solenoids		
	b)Pressure switches		
	c)Power pack Float , Thermal , clog switch.		

VIII.FIXTURE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for manual operation through push buttons.		
2	Check for auto operation through M codes.		
3	Check for timers,bypass bits, seat check,other related interlocks.		

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for direction of all coolant / conveyor motors.		
2	Check for cables/ hose / plugs termination & routing.		
3	Check the functions of following elements		
	a)Manual mode operations.(SKB push buttos, selector sw)		
	b)MDI / Auto mode operations		
	c)Float switch		
	b)Chip/ Screw Conveyor operation(lsw, proximity,,fwd/ rev etc)		
	c)HP coolant functions(clog,flow,air blast).		
	d)Push/Flush sequence through M180 command.		

VIII.ROTARY/ INDEX TABLES

SL.No	PARAMETERS	STATUS	REMARKS
1	Cable/ hoses are routed properly at safe position.		
2	Check for commanded position , table position & shortest path.		
3	Check for tool display & table position in case of index table.		

IX.CABINET & OPERATING PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the rating & spec.of following elements as per circuit diagram.		
	a)MPCB's		
	b)Contactors		
	c)MCB's		
	d)Transformer		
	e)PCB's,Relay's & relay boards		
2	Set all the MPCB's for 100% of rated current(ATC/Turret a.p. set valu		
3	Check the panel cooler with cab.door interlock.set for 35°C& lock par.		
4	Ensure the rotation of drive/heat sink fan & also if any other fans.		

CHECKED BY:-	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

CHECK LIST - ELECTRICAL

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5	Check for the following		
	a)Mounting of elements inside cabinet		
	b)Mounting of elements on operating panel		
	c)All Terminal tightness/ loose connections		
	d)Input/ output short connection		
	e)Ferruling of cable/wire(with adress if necessary)		
	f)Identification stickers		
	g)Moisture free beadings(both in cabinet & operating panel)		

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following		
	a)All emergency push buttons		
	b)Emergency monitor		
	c)Work lamp & m/c indication lamp(3 tier)		
	d)Auto door & m/c door limit switches if provided		
2	Check for - All earthing connections.		
3	Check for -Protective covers.		
4	Check for -PVC/Metal Channel cover fixing.		
5	Check for -Name plate & tags on all elements.		
6	Check for -Routing of cables & hoses.		
7	Check for- Model/Sl.No./ rating in M/C No. plate		
8	Mark the set point of MPCB with permanent marker.		

INTERNAL CHECKLIST POINTS

1) Safety Cover for CNC System	4) 15 Sticker R19 M	8) Screen missed on Lub motor
2) Pallet Stop box to be mount	5) AC mesh 5 Hole	* Air Hose
3) Coolant tags Example	6) Tube Color & Spare Gland Remove from TB9 JB)	

XI. PMC DATA

Instruction: Please note all D-bits which is enabled at the time of shipping .
Also give the details if it is used for special function

	7	6	5	4	3	2	1	0		7	6	5	4	3	2	1	0
D0	7	6	5	4	3	2	1	(0)	D13	7	6	5	4	3	2	1	0
D1	7	6	5	4	3	2	1	0	D14	7	6	5	4	3	2	1	0
D2	7	6	5	4	3	2	1	0	D15	7	6	5	4	3	2	1	(0)
D3	7	6	5	4	3	2	1	0	D16	7	6	5	4	3	2	1	0
D4	7	6	(5)	4	3	2	1	0	D17	7	6	5	4	3	2	1	0
D5	7	6	5	4	3	2	1	0	D18	7	6	5	4	3	2	1	0
D6	(7)	6	5	4	3	2	1	0	D19	7	6	5	4	3	2	1	0
D7	7	(6)	5	4	(3)	2	1	0	D20	7	6	5	4	3	2	1	0
D8	7	6	(5)	4	3	(2)	1	0	D21	7	6	5	4	3	2	1	0
D9	7	6	5	4	3	2	1	0	D22	7	6	5	4	3	2	1	0
D10	7	6	5	4	3	2	1	0	D23	7	6	5	4	3	2	1	0
D11	7	6	5	4	3	2	1	0	D24	7	6	5	4	3	2	1	0
D12	7	6	5	4	3	2	1	0	D25	7	6	5	4	3	2	1	0
D																	
D																	

CHECKED BY:-	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

CHECK LIST - ELECTRICAL

D110	00000000	1	1
D000	00000000	3	30

XII.PROGRAM/ LOGIC VERSION

Instruction: Please ckeck for latest version before making entry.

Logic version No.: DMA01-01-08.M0	Remarks: OPC
Macro version No.:	Remarks:

Progrm version Nos.:

O9001	A1.2.00	O9010	J1.A.00
O9002	B1.A.00	O9011	K1.A.00
O9003	C1.A.01	O9012	
		O9013	

XIII.CNC PARAMETERS

AXIS POSITION IN MACRO					
971	972	973	X- -350	Y- 67	Z- 172
991	992	993	X- -225	Y- 1.500	Y- -203.4

REFERENCE POSITION		X	Y	Z
2nd Ref.	1241	-328	1.50	
3rd Ref.	1242	-11	1.409.6	

RAPID TIME CONSTANT	X	Y	Z
1620	104	120	120
1621	22	96	120

	Check the following with system:		'1'		'0'	
	No.of Ladder steps**	Dgn:1141 # 6	8000		5000	**N/A for 0 iMD
	4th axis option**	1279 # 4 OR 1148 # 7	Yes		No	**N/A for 0 iMD
	Macro executer option	Dgn:1236 # 1	Yes		No	
	3111#7 = 0 (NPA) - Message display					

XIV.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

SL.No	PARAMETERS	REMARKS
1	Ensure the servo guide back up	
2	Check the various communication like RS232C,PCMCIA	
3	Edit title,& set time, date other original factory set values	
4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	
5	Take the back up of following :	
	a) CNC parameters	
	b) Pitch error compensation	
	c) PMC parameters	
	d) ID info	
	e) M-info	
	f) Ladder	
	g) Macro	
	h) S-RAM	
6	Make I/O channel '0' for RS232C communication	
7	Keep spare fuse inside the cabinet	
8	Ensure any updation/correction in electrical circuit diagrm	
9	Ensure all the elements are in good condition before shipping	
10	Compare CNC parametrers with standard data & store in backup file	

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ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

M/C NO.: M26-848	DATE: 07/09/12
MODEL: VMC Divider	CUSTOMER: SWASTIKA AUTOMATICS

M/C BACKUP (PATH) : \\arya2\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (20011-12)\FANUC 05.03

0i-MD ☐ 0i-MATE-MD ☑ ☐

PART No A02B-0321-B500

Sl.No.: E12551249 E12680347

SYSTEM OPTIONS:

β SERVO DRIVE SVPM

7.5i (20/20/40) ☐ 15i (40/40/80) ☑

PART No A06B-6164-H343

Sl.No.: [redacted] V1261502 H1580

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SPM

11i ☐ 15i ☐ 22i ☐ ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM2

40/40i ☐ 80/80i ☐ 160/160i ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM1(1'st)

40i ☐ 80i ☐ 160i ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM1(2'nd)

40i ☐ 80i ☐ 160i ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

FANUC I/O MODULE / I/O CARD

I/O MODULE WITH MPG ☐	I/O CARD WITH MPG ☐	I/O CARD WITHOUT MPG ☑
PARTNo.: A02B-0309-C001	PART No.: A20B-2002-0520	PART No.: A20B-2002-0521
Sl No. E12504834	Sl No.	Sl No.

AC REACTOR1

~~0.11mH/62A~~ ☐ 0.07mH/110A ☐ 0.13mH/46A ☑

PART No A81L-0001-0155

Sl.No.: [redacted] T12060477

CHECKED BY:- [signature]

VERIFIED BY:- [signature]

SERVO MOTOR - X AXIS

β8/3K	β12/3K	β22/3K	
☐	☑	☐	☐

PART No.: A06B-0075-B103 #100

Sl.No.: C125B3186 C126W1548

SERVO MOTOR - Y AXIS

β8/3K	β12/3K	β22/3K	
☑	☑	☐	☐

PART No.: A06B-0075-B103

Sl.No.: C125B3624 C126W13F8

SERVO MOTOR - Z AXIS

β12/3K	β22/3K	β22/2K	
☐	☑	☑	☐

PART No.: A06B-0082-B403

Sl.No.: C125A7~~2~~098 C126A655A

B AXIS DETAILS

N/A

ROTARY TABLE ☐ INDEX TABLE ☐ ☐

DRIVE:	βiSV-20	βiSV-40	
	☐	☐	

PART No.: A06B-____-H002

Sl.No.:

MOTOR:	β4/4K	β8/3K	
	☐	☐	

PART No.: A06B-0___-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

PMM AXIS DETAILS

N/A

SERVO ARM ☐ SERVO MAG ☐ TILTING TABLE ☐

DRIVE:	βiSV-20	βiSV-40	
	☐	☐	

PART No.: A06B-____-H002

Sl.No.:

MOTOR:	β4/4K	β8/3K	
	☐	☐	

PART No.: A06B-0___-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

SPECIAL INFORMATIONS:

CHECKED BY:- [illegible]

VERIFIED BY:- [illegible]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

ISOLATTION SWITCH (SW1) — AC 3 Ph 415V ☑

40A ☑ 63A ☐ 100A ☐

MAKE: SALZER ☑ ABB ☐

TYPE:

ISOLATTION TRANSFORMER (TFR1) PRIMARY- 415V/3Ph SECONDARY - 220V/3Ph

10KVA ☐ 12KVA ☑ 15.5KVA ☐ 11.5 KVA ☑

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 24130-08-12 4728-12-12

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A ☑ 32A ☐ 40A ☐ ☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

SECONDARY MCB (F19) — AC 3 Ph 220V / 3POLE ☑

40A ☑ 50A ☑ 63A ☐ ☐

MAKE: SCHNEIDER ☑ ABB ☐

TYPE:

DRIVE CONTROL MCB (F20) N/A — AC 3 Ph 220V / 2POLE ☐

1A ☐ 3A ☐ ☐ ☐

MAKE: SCHNEIDER ☐ MERLIN GERIN ☐

TYPE:

SPINDLE MOTOR FAN MPCB (F56) — AC 3 Ph 220V / 3POLE ☑

0.1-0.16A ☑ 0.25-0.4A ☐ ☐

MAKE: ABB ☑ SPRECHER &SCHU ☐

MPCB SETTING: 0.16A

PSM CONTACTOR (K19) — 2 30V AC ☑

CA3-30-10 ☐ A40-30-10 ☑ CA3- -N-11 ☐ ☐

MAKE: ABB ☑ SPRECHER &SCHU ☐

TYPE:

REGULATED POWER SUPPLY DETAILS

PS1 ☑ PS2 ☑ PS3 ☐

OUTPUT: 24V DC 10A ☑ 24V DC 20A ☐

MAKE: SCHNEIDER(ABL 1RPM24100) ☑ OMRON(S8VE 24024) ☐ KB ELECTORNICS ☐

Sl. No.: PS1: ABB CP-E 24/10 PS2: PS3:

ISOLATTION TRANSFORMER (TFR4) NA PRIMARY- 415V/1Ph SECONDARY - 200V/1Ph

900VA ☑ 1.5KVA ☐ ☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 1715-08-12

TFR4 PRIMARY MCB (F16) : 3A/2P ☑ NA ☐

MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

TFR4 SEC. MCB (F58) : 6A/1P ☑ ☐

MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

TFR4 SEC. MCB (F59) : 2A/1P ☑ 10A/1P ☑

MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

SPECIAL INFORMATIONS:

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

CONTROL CIRCUIT

SEALED KEY BOARD/MCP: Aishwarya Sl.No.: MP606-IT-1205009

TYPE: (YSL-AMS-SKB-2-REV1-10/09) ☐ YSL-SPARK-MCP2-REV2-05/05 ☐

ATC BRAKE RELAY (PCB 1) :

TYPE: 1 SSR RELAY -YSL-SSRP-BR-3-REV1-06/10 ☑ 2 SSR RELAY - YSL-S26-009-77-REV3-08/10 ☐

I/O DISTRIBUTON PCB (PCB 2) : Sl.No.:

TYPE: YSL-S26-009-56-REV0-04/08 ☑ YSL IFB- 50-FCC TB - REV3 01/06 ☐

CNC ON/OFF RELAY (PCB 3) :

TYPE: YSL -SSR-CNC ON / OFF-REV8-08/10 ☑

Z AXIS BRAKE RELAY (PCB 5) :

TYPE: YSL-SSRP-BR3-REV4-~~08/10~~ REV3-08/11 ☑

Z AXIS BRAKE CURRENT SENSINGRELAY (CSR1) :

TYPE: YSL ELECTRONICS (CSR1-2) ☑

AC RELAYS: Z BRAKE/EMG (ACR1) ☑ ATC/CAB. THERMAL(ACR2) ☑ I. TABLE THERMAL(ACR3) ☐

MAKE: OMRON ☑ ELESTA ☐

RELAY BOARD(STD-X20) 1RELAY(ELECTRO MAGNETIC-DC) ☑

TYPE: YSL-UMK-1RM-24V- DC-CN-FI-2-REV1-12/09 ☑

RELAY BOARD(STD-X31) 1RELAY(ELECTRO MAGNETIC-DC/AC) ☑

TYPE: YSL-S26-009-082-REV1-06/10 ☑

RELAY BOARD(APC-X25) 8 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-064-~~REV3-08/08~~ REV4-09/11 ☑

RELAY BOARD(FIXTURE-X26) 2 / 4 / 8 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-062-REV3-08/08 ☐ S26-009-063-REV3-08/08 ☐ S26-009-064-REV3-08/08 ☐

RELAY BOARD(TOOL DECLAMP - X33) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

RELAY BOARD(- X) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

EMERGENCY CONTACTOR (K22) 24V DC ☑

CS8C-40E ☐	KC6-40E ☑	MAKE: ABB ☑ SPRECHER &SCHU ☐
		TYPE:

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MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

POWER FAILURE DETECTION MODULE (PFDM) : 24V DC ☐

MAKE: YSL-3PFMR-2 ☐

CAPACITOR BANK (DP1) : N/A 24V DC ☑

MAKE: YSL-PF-CHU-1 ☑

SINGLE PHASE POWER SUPPLY CIRCUIT

MCB (F18) N/A AC 1 Ph 230V / 1POLE ☑

10A ☑ ☐ ☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

PANEL COOLER: POWER: 500W ☐ 1000W ☑

MAKE: ADVANCE (TURBO1000R33) ☑ ADVANCE (TURBO500R50) ☐ ACE (IGLOO - 750) ☐

Sl.No.:

UTILITIES (CABINET):

CABINET LAMP: PEARL(CFL-11W) ☑ PHILIPS(CFL-11W) ☐ ☐

GANG BOX: 2 SWITCHES + 1SOCKET + I INDICATOR ☐

DOOR INTERLOCK: LIMIT SW (HONEYWELL-ZLD A01D) ☑

MCB FOR CAB. LAMP, 5 PIN SOCKET (F66-3A/1PMAKE: SCHNEIDER) ☑

UTILITIES (SWITCHGEARS):

ADD ON BLOCK(PCB5/PSM(K19) INTERLOCK): SPC&SCH(CS7-PV) ☐ SPC&SCH(CA3-P) ☐

ADD ON BLOCK(STAR DELTA INTERLOCK): SPC&SCH(CS7-PV) ☐ SPC&SCH(CA3-P) ☐

ADD ON BLOCK(HYDRAULIC FAN POWER): (KQ) SPC&SCH(CS8-P22Z) ☐ ABB(CAF6-20-M) ☑

ADD ON BLOCK(AUTO DRAIN SOL): (KW) SPC&SCH(CS8-P22Z) ☐ ABB(CAF6-20-M) ☑

SPINDLE STAR / DELTA CIRCUIT

CONTACTORS (K20 / K21) N/A 230V AC ☐

CA3-72N-L11 ☐ CA7 - 85 ☐ ☐

MAKE: ABB ☐ SPRECHER &SCHU ☐

TYPE:

RELAY BOARD(X32) 2 RELAYS (SSR TYPE) ☐

TYPE: S26-009-80-CN--F-24DC-230AC ☐

SPECIAL INFORMATIONS:

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

MAGAZINE MOTOR POWER

MPCB (F1)	1-1.6A	☑	☐	CONTACTOR (K1/K2)	CA8-9C	☐	VBC6-30-01 ☑
MPCB SETTING:	1A	☑	☐	MOTOR POWER:	1.1KVA/1.5A	☐	1.6KVA/2.3A ☑
SOLID STATE RELAY (SSR1):				MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST)			☑

ARM MOTOR POWER N/A

MPCB (F2)	1-1.6A	☐	☐	CONTACTOR (K3/K76)	CA8-9C	☐	VBC6-30-01 ☐
MPCB SETTING:	1A	☐	☐	MOTOR POWER:	1.1KVA/1.5A	☐	2.3KVA/3.2A ☐
SOLID STATE RELAY (SSR3):				MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST)			☐

LUBRICATION MOTOR POWER

MCB (F3)	1A/3P ☑	MPCB(0.63-1A) ☐		CONTACTOR (K4)	CA8-9C	☐	BC6-30-01 ☑
				MOTOR POWER:	0.09KW/0.5A	☐	0.12KW/0.6A ☑

FLOOD COOLANT MOTOR POWER

MPCB (F4)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K5)	CA8-9C	☐	BC6-30-01 ☑
MPCB SETTING:	0.9A ☑	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☑		1.5KW/3.4A ☐
SURGE SUPRESSOR(RC4):				YSL-NIR-RC3-1/022-575VAC			☑

CHIP PUSH COOLANT MOTOR POWER

MPCB (F5)	0.63-1A ☐	2.5-4A ☑		CONTACTOR (K6)	CA8-9C	☐	BC6-30-01 ☑
MPCB SETTING:	0.9A ☐	3.4A ☑		MOTOR POWER:	0.37KW/0.9A ☐		1.5KW/3.4A ☑
SURGE SUPRESSOR(RC5): YSL-NIR-RC3-1/022-575VAC ☑				COMMON CONTACOR KIT - PUSH/FLUSH(ABB-VBC6)			☑

CHIP FLUSH COOLANT MOTOR POWER

MPCB (F6)	0.63-1A ☐	2.5-4A ☐		CONTACTOR (K7)	CA8-9C	☐	BC6-30-01 ☑
MPCB SETTING:	0.9A ☐	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☐		1.5KW/3.4A ☑
SURGE SUPRESSOR(RC6): YSL-NIR-RC3-1/022-575VAC ☑				COMMON MPCB WITH CHIP PUSH			☑

COOLANT GUN MOTOR POWER

MPCB (F27)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K25)	CA8-9C	☐	BC6-30-01 ☑
MPCB SETTING:	0.9A ☑	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☑		1.5KW/3.4A ☐
SURGE SUPRESSOR(RC27):				YSL-NIR-RC3-1/022-575VAC			☑

HIGH PRESSURE COOLANT MOTOR POWER

MPCB (F9)	4-6.3A ☐	2.5-4A ☐		CONTACTOR (K10)	CA8-12C	☐	BC7-30-01 ☐
MPCB SETTING:	4.9A ☐	5.2A ☐		MOTOR POWER:	2.2KW/4.9A ☐		2.2KW/5.2A ☐
SURGE SUPRESSOR(RC9):				YSL-NIR-RC3-1/022-575VAC			☐

HYDRAULIC 1 MOTOR POWER

MPCB (F8)	4-6.3A ☐	2.5-4A ☑		CONTACTOR (K9)	CA8-9C	☐	BC6-30-01 ☑
MPCB SETTING:	2.7A ☑	3.5A ☐		MOTOR POWER:	1.1KW/2.7A ☑		1.5KW/3.5A ☐
SURGE SUPRESSOR(RC8):				YSL-NIR-RC3-1/022-575VAC			☑

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VERIFIED BY:- [signature]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

INDEX TABLE MOTOR POWER

MPCB (F12)	2.5-4A ☐	☐	CONTACTOR(K15/K16)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	1.6KVA/2.3A ☐
SOLID STATE RELAY (SSR5):			MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☐		

CHIP CONVEYOR 1 MOTOR POWER

MPCB (F10)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K11/K12)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC10):			YSL-NIR-RC3-1/022-575VAC ☐		

CHIP CONVEYOR 2 MOTOR POWER

MPCB (F11)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K13/K14)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC11):			YSL-NIR-RC3-1/022-575VAC ☐		

CONVEYOR DRUM WASH MOTOR POWER

MPCB (F7)	0.63-1A ☐	2.5-4A ☐	CONTACTOR (K8)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	0.9A ☐	3.4A ☐	MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☐
SURGE SUPRESSOR(RC7):			YSL-NIR-RC3-1/022-575VAC ☐		

SCREW CONVEYOR 1 MOTOR POWER

MPCB (F13)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K17/K18)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	1.4A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC13):			YSL-NIR-RC3-1/022-575VAC ☐		

SCREW CONVEYOR 2 MOTOR POWER

MPCB (F30)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K28/K29)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	1.4A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC30):			YSL-NIR-RC3-1/022-575VAC ☐		

MOTOR POWER (OPTIONAL-1)

MPCB (F)	☐	☐	CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC ☐		

MOTOR POWER (OPTIONAL - 2)

MPCB (F)	☐	☐	CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC ☐		

SPECIAL INFORMATIONS: MPCB'S- SPRECHER &SCHU(KTA7-25S) ☐ ABB (MS 116) ☑

CHECKED BY:-

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

ATC & SPINDLE DETAILS

SPINDLE MOTOR: MODEL:	β3 ☐	β6 ☑	β12 ☐	☐	
MOTOR RPM:	6000 ☐	8000 ☐	10000 ☐	☐	
MOTOR POWER:	5.5/3.7 KW				
PART No:	A06B-[illegible]				
Sl. No:	[illegible] C126J01A				
GEAR RATIO:	1:1 ☐	1:1.25 ☐	1:1.5 ☑	DIRECT DRIVE ☐	
POWER LIMIT:	IN KW:				
SPEED LIMIT:	IN RPM: 60-6000 Rpm				

PROXIMITY SENSORS-SPINDLE

MAKE; TEKNIC ☑ | TYPE: (24V DC - PNP) ☑

SPINDLE ORIENTATION (PXS1) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☑	☐	☑	☐	☐
TOOL CLAMP(PXS3) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
TOOL DECLAMP(PXS4) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☑	☐	☐
NO TOOL IN SPINDLE(PXS32) ☑	M08X50	M08X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☑	☐	☑	☐	☐
ONE ROTATION(PXS39) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☑	☐	☐

PROXIMITY SENSORS-ATC(DISC TYPE)

MAKE; TEKNIC ☑ | TYPE: (24V DC - PNP) ☑

MAGAZINE IN POSITION (PXS2) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
MAGAZINE REFERENCE (PXS46) ☐	M12X73	FG2X05APO24-5000LFLAT		
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐		☐

PROXIMITY SENSORS-ATC(TWIN ARM TYPE)

MAKE; TEKNIC ☐	TYPE:EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☐
MAGAZINE IN POSITION PROXIMITY (PXS2) ☐	MAG REFERENCE PROXIMITY (PXS46) ☐
POCKET UP PROXIMITY (PXS5) ☐	POCKET DOWN PROXIMITY (PXS6) ☐
ARM POSITION 1 PROXIMITY-PS1 (PXS7) ☐	☐
ARM POSITION 2 PROXIMITY-PS2 (PXS58) ☐	ARM POSITION 3 PROXIMITY-PS3 (PXS59) ☐

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PNUEMATIC ELEMENTS (SPINDLE + ATC)

MAKE; SUCO ☐	TYPE:(24V DC - NO CONTACT) ☐
MAIN AIR PRESSURE SWITCH (PRS 2) ☐	SPINDLE AIR PRESSURE SWITCH (PRS 101) ☐
AIR PURGE PRESSURE SWITCH (PRS 102) ☐	☐
MAKE; FESTO ☐ SMC ☐	TYPE:(24V DC -COIL) ☐
SPINDLE AIR JET SOLENOID (Y1) ☐	SPI AIR PURGE SOLENOID (Y115) ☐
POCKET UP SOLENOID (Y2) ☐	POCKET DOWN SOLENOID (Y3) ☐
TOOL CLAMP SOLENOID (Y51) ☐	TOOL DECLAMP SOLENOID (Y4) ☐
AIR THROUGH NOZZLE SOLENOID (Y99) ☐	☐

HYDRAULIC ELEMENTS (SPINDLE + ATC)

MAKE; YUKEN ☐ REXROTH ☐	TYPE:(24V DC -COIL) ☐
☐	TOOL DECLAMP SOLENOID (Y4) ☐

UTILITIES(SPINDLE + ATC)

TOOL CHANGE POSITION LIMIT SWITCH- (LS3) :	TEKNIC SG-01-R-610 ☐	☐
TOOL DECLAMP PUSH BUTTON :	ILLUMINATED GREEN ☐	☐
ATC / I.TABLE DISPLAY SEL SWITCH :	2 POSITION ☐	☐
MAG TEST MODE KEY SWITCH (KSW7) :	2 POSITION ☐	☐
	☐	☐

SPECIAL INFORMATIONS:

AUTO DOOR & CYCLE START DETAILS

MAKE; FESTO ☐	TYPE:(24V DC -COIL) ☐
AUTO DOOR OPEN SOLENOID (Y6) ☐	AUTO DOOR CLOSE SOLENOID (Y7) ☐
AUTO DOOR SAFETYSOLENOID (Y8) ☐	☐
DOOR SWITCHES MAKE; TECNIK ☐ FESTO ☐	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT) ☐ READ SWITCHES (RSW30& RSW31) ☐
CYCLE START SWITCH MAKE; ☐ TELEMECHANIC ☐	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT) ☐ ☐
DOUBLE HAND CYCLE START BOX MAKE; TECNIK ☐ ☐	CYCLE START PB (MUSHROOM HEAD GREEN) (02 Nos) ☐ EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) ☐
AUTO DOOR OPERATION MAKE; TECNIK ☐ ☐	AUTO DOOR OPEN/CLOSE PB (GREEN) ☐ AUTO DOOR INTERLOCK KEY SW (2 POSITION) ☐

SPECIAL INFORMATIONS:

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COOLANT TANK / CONVEYORS DETAIL

UTILITIES (STD COOLANT)					
COOLANT LEVEL FLOAT SWITCH (PXS 8) :	FILPRO(LIM 18PF SC)	☑		☐	
COOLANT GUN SEL SW :	2 POSITION	☑		☐	
FLUSH COOLANT VALVE (Y128) :	AVCON	☐		☐	
COOLANT PLUG (HC1) :	16 PIN	☑		24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐		RHS	☑
UTILITIES (HP COOLANT):					
FLOW SWITCH (FLW1) :	IFM (SH0 500)	☐		☐	
CLOG SWITCH (FCLG1) :	HIRLEKAR	☐		SALVIN	☐
AIR BLAST SOLENOID (Y5) :	FESTO	☐		☐	
M/C DOOR CLOSE LIMIT SW (LS5) :	TEKNIC(NG1 RS 510)	☐		☐	
UTILITIES (CHIP CONVEYOR):					
CHIP CONVEYOR 1 O/L LIMIT SW (LS12) :	TEKNIC(NG1 RS 510)	☐		☐	
CHIP CONVEYOR 2 O/L LIMIT SW (LS13) :	TEKNIC(NG1 RS 510)	☐		☐	
CHIP CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐	
CHIP CONVEYOR PLUG (HC3) :	10 PIN	☐		24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐		RHS	☐
UTILITIES (SCREW CONVEYOR):					
SCREW CONVEYOR 1 O/L PROXIMITY (PXS20) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)		☐
SCREW CONVEYOR 2 O/L PROXIMITY(PXS104) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)		☐
SCREW CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐	COMMON WITH CHIP.CNYR		☐

SPECIAL INFORMATIONS:

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) :	SHRIDHAN	☐		☐	
OIL CLOG SWITCH (CLG2) :	HYDAC	☐		☐	
HYDRAULIC THERMAL SWITCH (TH4) :	CARE SYSTEM	☐		☐	
HYDRAULIC PLUG (HC2) :	16 PIN	☑		24 PIN	☐
MAKE;INDO ELECTRIC ☐ LOCATION(FROM M/C REAR) :	LHS	☑		RHS	☐

SPECIAL INFORMATIONS:

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LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH (FL1) :	CENLUB	☑		☐
LUB PRESSURE SWITCH (PRS1) :	SUCO	☑		☐
AUTO DRAIN SOLENOID (Y44) :	ROTEX	☑		☐

MACHINE LAMPS

MAKE:			PATANKAR	☐	AISHWARYA	☐
MACHINE LAMP1 (ML1) :	11W	☐	9W+9W	☑	36W	☐
MACHINE LAMP2 (ML2) :	11W	☐	9W+9W	☑	36W	☐
MACHINE LAMP3 (ML3) :	11W	☐	9W+9W	☐	36W	☐
MAHINE INDICATION LAMP(PL5) :	1 TIER	☐	3 TIER	☑	3 TIER WITH HOOTER	☐
MAKE:			PATANKAR	☑	LAXMI(VANTECH	☐

UTILITIES(MAHINE)

EMERGENCY MONITOR (PCB 12)			YSL-EMG-PCB-4-REV2-06/10	☑
HAND HELD MPG:		FANUC	☐	FUTURE(TAIWAN) ☐
		Sl. No.:		
TPM TRACK :		PIONEER CT	☐	☐
		Sl. No.:		
SERVO VOLTAGE STABILIZER:		15KVA	☐	25KVA ☐
INPUT:	OUTPUT:	MAKE: SERVOMAX	☐	☐
		Sl. No.:		
COOLANT / SPINDLE CHILLER UNIT:		MAKE:	POINT ☐	☐
POWER:	MCB(F31):4A / 3P	Sl. No.:		
CONNECTOR :	INDO ELECTRIC (10 PIN) ☐	LOCATION:	LHS ☐	RHS ☐

FIXTURE DETAIL

10A

MAKE;	☐	PNEUMATIC ☐	HYDRAULIC ☐
STATION 1 CLAMP SOLENOID	(Y70) ☐	STATION 1 DECLAMP SOLENOID	(Y71) ☐
STATION 2 CLAMP SOLENOID	(Y72) ☐	STATION 2 DECLAMP SOLENOID	(Y73) ☐
STATION 3 CLAMP SOLENOID	(Y84) ☐	STATION 3 DECLAMP SOLENOID	(Y85) ☐
STATION 4 CLAMP SOLENOID	(Y86) ☐	STATION 4 DECLAMP SOLENOID	(Y87) ☐
MAKE;	☐	PNEUMATIC ☐	HYDRAULIC ☐
STATION 1 CLAMP PRESSURE SW	(PRS20) ☐	STATION 2 CLAMP PRESSURE SW	(PRS22) ☐
STATION 3 CLAMP PRESSURE SW	(PRS31) ☐	STATION 4 CLAMP PRESSURE SW	(PRS32) ☐

SPECIAL INFORMATIONS: CLAMP PB'S (GREEN) -

DECLAMP PB'S (RED) -

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APC (LINEAR TYPE) DETAIL

MANUAL OPERATION PANEL :	YSL-AMS-APC-PCB-REV0-07/06 (REV1-06/07)	☑
PALLET SET UP PB :	LUMINOUS GREEN	☑
EMERGENCY PB :	MUSHROOM HEAD RED	☑

PROXIMITY SENSORS:

MAKE; TEKNIC ☐ TYPE: (24V DC - PNP) ☐

PALLET RECEIVED PROXIMITY(PXS11) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☑	
ARM 1 IN PROXIMITY (PXS13) ☑	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑	
ARM 1 OUT PROXIMITY (PXS14) ☑	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑	
ARM 2 IN PROXIMITY (PXS15) ☑	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 2 OUT PROXIMITY (PXS16) ☑	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑	

PNEUMATIC SOLENOIDS:

MAKE; FESTO ☐	TYPE: 24V DC -COIL ☑
APC DOOR OPEN SOLENOID (Y12) ☐	APC DOOR CLOSE SOLENOID (Y13) ☑
ALLET PAD CLEAN SOLENOID (Y18) ☐	☐

HYDRAULIC SOLENOIDS:

MAKE; YUKEN ☐ REXROTH ☐	TYPE: 24V DC -COIL ☑
PALLET CLAMP SOLENOID (Y10) ☐	PALLET DECLAMP SOLENOID (Y11) ☑
ARM 1 IN SOLENOID (Y14) ☐	ARM 1 OUT SOLENOID (Y15) ☑
ARM 2 IN SOLENOID (Y16) ☐	ARM 2 OUT SOLENOID (Y17) ☑

HYDRAULIC PRESSURE SWITCHES:

MAKE; POLYHYDRON ☐	TYPE: ☑
PALLET CLAMP PRSW - (PRS3) ☐	PALLET DECLAMP PRSW (PRS60) ☑

LIMIT SWITCHES:

MAKE; TEKNIC(NG1VS 510) ☐	JAI BALAJI (JLSSPR) ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☐	PALLET 2 ON STATION LIMIT SW (LS11) ☑

READ SWITCHES:

MAKE; FESTO ☐	APC DOOR INTERMEDIATE READ SW (RSW 65) ☐
APC DOOR OPEN READ SW (RSW 58) ☐	APC DOOR CLOSE READ SW (RSW 59) ☑

SPECIAL INFORMATIONS:

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FANUC Series 0i Mate-M Model D DATA SHEET

得意先名: FANUC INDIA PRIVATE LIMITED
CUSTOMER: YD79733- 000
CONTRACT NO.: A02B-0321-B500
MODEL NO.: E126B0347
SERIAL NO.:
SETTING OF INPUT:

DATE OF MANUFACTURE: 2012-06
DATE OF TEST : 2012/06/06
CHECKED BY : A. Koike
APPROVED BY : K.Miura

CNC IDENTIFICATION NUMBER

3C7B5D01
6A916D86
5E34C389
4A92C4EC

EDITION OF PRINTED BOARD & ROM

NAME	EDIT
A20B-8200-0849	01A

NAME	EDIT
A02B-0319-H500#60W3	05
A02B-0319-H580#40B0	06
A02B-0319-J541#60VM	06
A02B-0319-J571#658E	05
A02B-0319-J572#658F	04
A02B-0319-J579#BW50	03
A02B-0320-J578#DX50	03
A02B-0320-J578#DX51	03
A02B-0322-H501#D5F1	23
A06B-6057-H507#90C5	06

NAME	EDIT

LIST OF CARD , DIMM & BACK BOARD

A20B-3300-0638	1
A20B-3900-0286	1
A20B-8002-0703	1
A20B-8200-0560	1

本データシートはファナック工場出荷時のCNC装置の装置情報です
Information contained in this Data Sheet is system configuratio
at the time of FANUC factory shipment.

NAME	NAME	SPECIFICATION	ORDR
BASIC CONTROL UNIT	ベーシック ユニット	A02B-0321-B500	1
AXIS CONTROL CARD A2 FROM/SRAM MEMORY A1 MAIN BOARD POWER SUPPLAY UNIT DISPLAY UNIT NAME PLATE	ジク セイギョ カード A2 FROM/SRAMメモリ A1 メインボード デンゲン ユニット ヒョウジ ユニット メイバン	A02B-0319-H032 A02B-0319-H051 A02B-0319-H105 A02B-0319-H111 A02B-0319-H144#M A02B-0322-H190	1 1 1 1 1 1
PACKAGE 3	パッケージ 3	A02B-0322-H532	1
I/O UNIT FOR Oi *MANUAL PULSE GENERATOR	I/O UNIT *シュドウ パルス ハッセイ	A02B-0309-C001 A860-0203-T001	1 1
	SPINDLE AMP.	A06B-6164-H343#H580	1
	BIS12/3000 BIS12/3000 BIS22B/3000 SPB6/10000I	A06B-0078-B103 A06B-0078-B103#0100 A06B-0082-B403 A06B-1445-B103	1 1 1 1

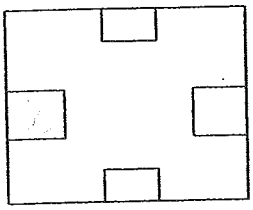
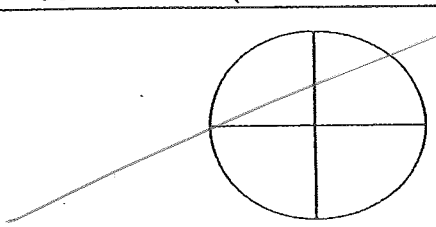
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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

MACHINE No. : M26-848	CUSTOMER : M/S Swastika Auto

I. Motor Mounting Bracket Face Out and CONCENTRICITY for Direct Drive

	FACE OUT (Recommended ≤ 0.015 mm)	CONCENTRICITY (Recommended ≤ 0.010 mm)
		 N/A

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
1	Spindle Connector Radial Run Out	≤ 0.010 mm	
2	Motor Current at Maximum RPM	β3 < 15%, β6 / α6 & ABOVE <10%	
3	CTS Rod Guide Radial Run Out (DDCTS)	≤ 0.010 mm	
4	Spindle Clamping Force	As per DS-DSN-PROD-01	
5	Spindle Motor Specification	As per Work Order	
II	**INDIRECT DRIVE**		
1	Motor Pulley Radial Run Out	≤ 0.050 mm	30µ
2	Spindle Pulley Radial Run Out	≤ 0.050 mm	30µ
3 a)	Belt Tension	As per DS-DSN-PROD-02	210
b)	Belt Specification	Size : 620	Quantity : 1
4	Motor Current at Maximum RPM	β3 < 20%, β6 / α6 & ABOVE <15%	12%
5	Rotary Joint Radial Run Out (ID CTS)	≤ 0.040 mm	N/A
6	Spindle Clamping Force	As per DS-DSN-PROD-01	Good
7	Spindle Motor Specification	As per Work Order	B3
8	Pulley Ratio	As per Work Order	1 : 1.5
III	Axis Dead Stopper Value / Butt Referencing Value (As per Drawing)		

	Axis	MOTOR END	BRAKE END
1	X	17.700	-541.800
2	Y	-418.852	23.100
3	Z	178.728	-377.412

IV. % of axis current drawn (Recommended as per Data sheet DS -QA -02)

	Axis	100 % Rapid rate
	X	33%
	Y	41%
	Z Down	29%
	Z Up	48%

V	Accumulator Oil Pressure	As per Circuit Diagram	N/A
	Accumulator Gas Pressure	As per Circuit Diagram	N/A
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S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
VI	**ATC ALIGNMENT (TURRET TYPE)**		
1	Parallelism of gripper Arm face wrto. X - Axis	≤ 0.050 mm	30µ
2	Tool Ejection Stroke	As per DS - DSN-PROD-01	0.84mm
3	Swing Lever Gap	As per Drawing	8.1mm
4	Side Cam Alignment	0.020 mm Maximum	10µ
5	Gap between Draw Bar bush & Roller/Screw	1 ± 0.1 mm	0.8mm
VII	**ATC ALIGNMENT (TWIN ARM)**		
1	X - Axis	≤ 0.2 mm	
2	Y - Axis	≤ 0.2 mm	
3	Z - Axis	≤ 0.05 mm	
4	Twin Arm bottom face parallelism in Y-Axis	≤ 0.1 mm	
5	Gap between Draw Bar Bush & Roller	1 ± 0.1 mm	
6	Tool Ejection stroke	As per DS - DSN-PROD-01	
7	Gap between deflection Pads	0.02 TO 0.050 mm	
VIII	**APC ALIGNMENT - LINEAR TYPE**		
1	Height difference of rest pads	≤ 0.010 mm	
2	Pallet change Time	As per Work Order	
IX	**APC ALIGNMENT- ROTARY TYPE**		
1	APC Base parallelism w.r.to. X - Axis	≤ 0.1 mm	
	APC Base parallelism w.r.to. Y - Axis	≤ 0.1 mm	
2	Arm Top parallelism w.r.to. X - Axis	≤ 0.05 mm	
	Arm Top parallelism w.r.to. Y - Axis	≤ 0.05 mm	
3	Arm 1 Alignment ' FACE ' w.r.to X- Axis	≤ 0.02 mm	
	Arm 1 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' FACE ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
4	Offset between Arm 1 & Arm 2 - Face	≤ 0.030 mm	
	Offset between Arm 1 & Arm 2 - Taper	≤ 0.030 mm	
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S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
5	Pallet - 1 Wedge Alignment along X - Axis	≤ 0.020 mm	
	Pallet - 2 Wedge Alignment along X - Axis	≤ 0.020 mm	
6	Gap between Pallet -1 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -1 side & Arm along Z - Axis	0.1 ± 0.02 mm	
	Gap between Pallet -2 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -2 side & Arm along Z - Axis	0.1 ± 0.02 mm	
7	Pallet Sagging - Pallet -1 'W.r.to. X & Y - Axis	≤ 1 mm	
	Pallet Sagging - Pallet -2 'W.r.to. X & Y - Axis	≤ 1 mm	
8	Pallet Height Variation	As per TEST CHART	
9	Backlash between Rack & Pinion at distance of 500 mm	≤ 1 mm	
10	Pallet Change Time	As per Work Order	

S.No.	CHECK POINTS	REMARKS
A	**AXIS**	
1	Ensure geometrical accuracies as per test chart.	ok
2	Ensure plugs are fitted for LM guide ways.	N/A
3	Ensure butt reference function.	OK
4	Ensure effective wiping of saddle cover wipers.	ok
5	Ensure lube timer is set 12 min for HG and 20 min for LM Guide ways.	
6	Ensure leveling bolt locknuts are tighten after machine leveling.	ok
7	Check the all axes reference arrows are matched & Direction name plate fixed.	
8	Ensure tightness & marking of all screws with Yellow paint marker	ok
9	Ensure proper setting of Limit switches.	N/A
B	**ATC**	
1	Check oil level is maximum in ATC gear box.	ok
2	Check tool change position limit switch for proper functioning & band width (Max. 1 mm on each side).	N/A
3	Ensure smooth function of Tool change without abnormal/unhealthy noise.	ok
4	Ensure ATC bracket & Magazine Dowelling.	

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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

S.No.	CHECK POINTS	REMARKS
C	**SPINDLE**	
1	Check the tightness of push stud.	ok
2	Ensure air jet through all the (4 / 8) holes at oriented spindle position	ok
3	Ensure supply of air to spindle Air Purge system (If provided)	NA
4	Ensure proper setting of proximities.	ok
5	Ensure smooth running of spindle without abnormal/unhealthy noise	ok
6	Ensure dummy plate fixing on CTS slot in Non CTS machine.	NA
7	Check the Tool not in spindle proximity message.	ok
D	**COOLANT**	
1	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe	ok
2	Ensure proper working of float switch (by shifting the sensor upwards).	
3	Ensure all the hoses, fittings tightened and routed properly.	ok
4	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).	
5	Ensure CTS flow sw.setting by opening the bag filter air vent ball valve.(Optional)	
6	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (If provided)	
7	Ensure CTS pin locator hole is plugged (If provided)	
8	Ensure that twin arm plunger & CTS pin locator screw do not foul (if provided)	
9	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)	NA
10	Ensure High pressure air blast after CTS switched off (If provided)	
11	Ensure drum wash in conveyors (If provided)	
12	Check conveyor overload message & Direction (If provided)	
13	Check oil in conveyor gear box. (if provided)	
14	Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).	
15	Ensure splash guard fixed to conveyor.	
E	**APC**	
1	Ensure proper setting of proximities.	ok
2	Ensure supply of air jet to pallet rest pad cleaning.	ok
3	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise	ok
4	Ensure APC base Dowelling.	ok
5	Check pallet initialization in APC Machine	
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S.No.	CHECK POINTS	
F	**HYDRAULIC, PNEUMATIC & LUBRICATION**	
1	Ensure pneumatic main line pressure switch setting (4 bar) & error message are satisfactory	
2	Ensure Spindle air jet pressure switch setting (3 bar) & error message are satisfactory	
3	Ensure Spindle air purge pressure switch setting (1.2 bar) & error message are satisfactory	NO
4	Ensure proper setting of oil flow (1 drop/tool change) in FRL unit. & Oil Level low message	
5	Ensure proper routing of FRL drain hose.	ok
6	Ensure identification name plates are fixed on power pack hoses.	ok
7	Ensure Ball valve fixed to FRL Unit inlet	ok
8	Ensure pneumatic quick change coupling fixed to 4th axis provision	—
9	Ensure no oil leakage in hydraulic valves & fittings	
10	Ensure hydraulic flow control valve grub locking in APC machine.	
11	Ensure pneumatic counter balance bracket Dowelling.	NA
G	**GUARDING ASSEMBLY**	
1	Ensure all internal guards are assembled.	ok
2	Ensure smooth movement of sliding door without play and unhealthy noise	ok
3	Ensure Nylock nuts are fixed to door rollers.	ok
4	Ensure ATC door LM rail plugs fixed.	NA
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	NA
H	**ANY OTHER POINTS**	
NAME & SIGN	CHECKED BY / APPROVED BY	

ACE MANUFACTURING SYSTEMS LTD

GEOMETRICAL TEST CHART(VMC / CMV)

AMS — In Pursuit of Excellence

PERMISSIBLE TOLERANCE AS PER TEST CHART (DS-DSN-PROD-TC-01)

1. SPIRIT LEVEL READING:

Along X-Axis			Along Y-Axis		
0	0	0	1/2	0	0
0	0	0	0	0	0
1/2	0	0	0	0	0

NO - ~~002~~ 6862

2. TABLE TOP READINGS:

Along X-Axis	Along Y-Axis

PALLET - II

Along X-Axis	Along Y-Axis

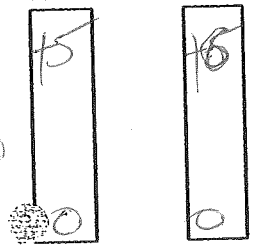
3. T SLOT READING:

0	+10

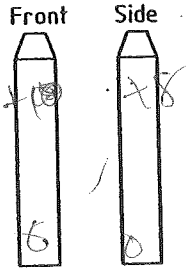
PALLET - II

0	+12

4. COLUMN READINGS :



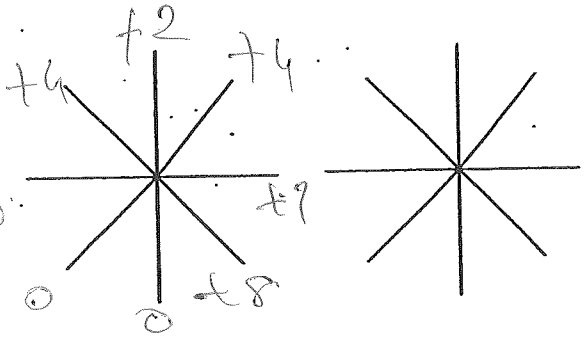
5. MANDREL READINGS:



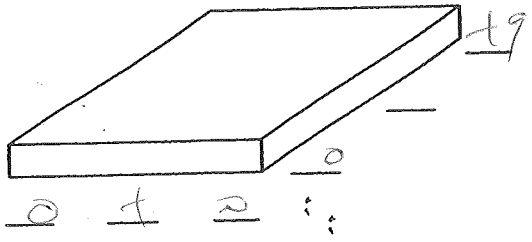
6. TURN AROUND READING:

RUNOUT:

a. TOP :
   1. 29
b. BOTTOM :
   1. 6
   2. 8
   3. 10
   4. 9



7. SQUARENESS READING (X&Y):



8. PALLET HEIGHT VARIATION:

Pallet-1:

Pallet-2 :

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			6....	F-QA-FA-05-R1

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Sign: Date: 21/3/12

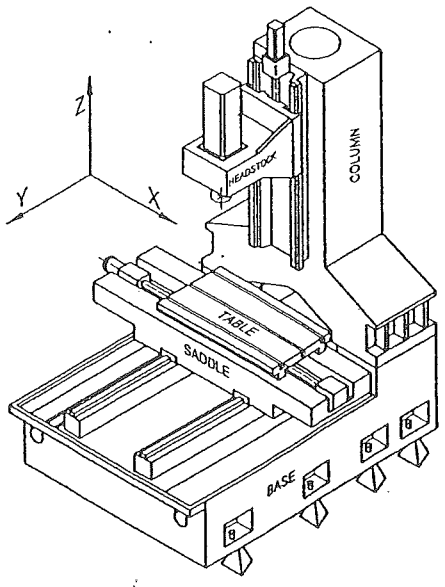
ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

PRODUCT	:	VERTICAL MACHINING CENTER (FIXED COLUMN)
MODEL NAME	:	WINNER
MACHINE NUMBER	:	M26 – 848
DATE OF TESTING	:	19/12/12



CHECKED BY
QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

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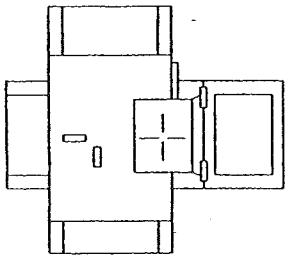
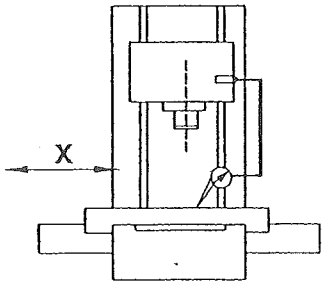
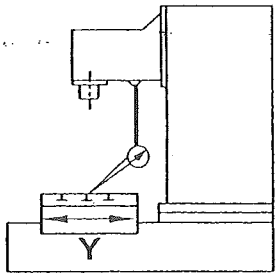
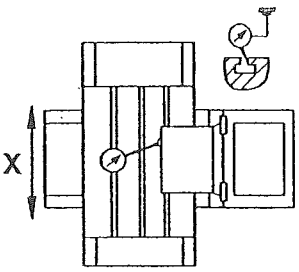
Sign: Date: 21/3/12

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DESIGN VERIFICATION / VALIDATION PLAN CHART

AMS — In Pursuit of Excellence

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm — Tolerance: Permissible	Tolerance: Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / m ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0.005 LEVEL ALONG Y-AXIS : 0.005
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : PALLET 1 : 0.012 PALLET 2 : 0.013 (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : PALLET 1 : 0.013 PALLET 2 : 0.015 (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : PALLET 1 : 0.009 PALLET 2 : 0.018 (FOR APC MACHINE ONLY)

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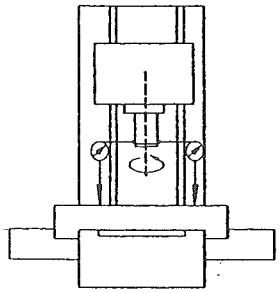
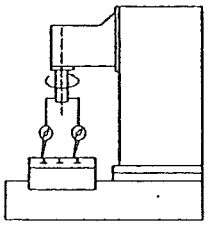
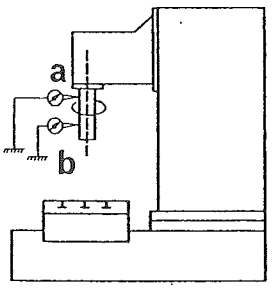
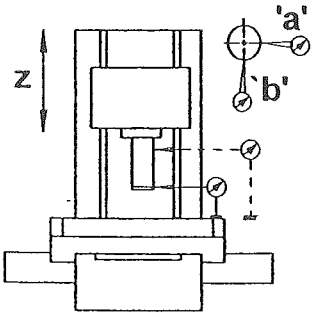
Sign: Date: 21/3/12

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DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : PALLET 1 : 0.010 PALLET 2 : 0.012 (FOR APC MACHINE ONLY)
G6	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : PALLET 1 : 0.010 PALLET 2 : 0.008 (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR SPINDLE NOSE b) AT A DISTANCE OF 300 FROM SPINDLE NOSE		a. 0.01 NEAR SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 0.002 b. 0.016
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 0.004 b. 0.008

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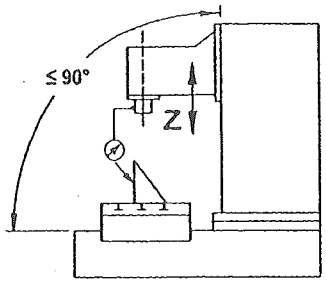
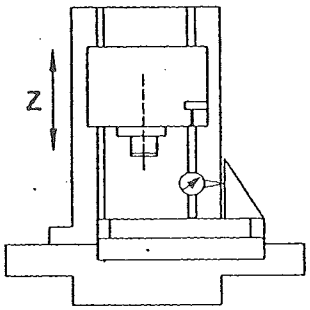
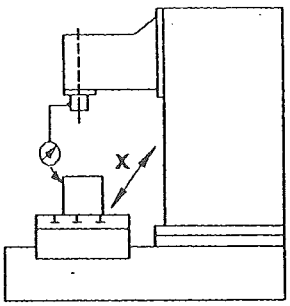
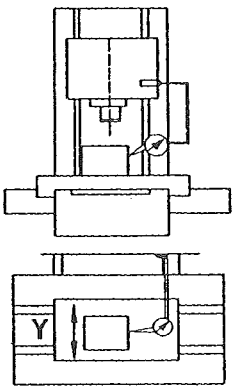
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ACE MANUFACTURING SYSTEMS LTD

AMS In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN YZ PLANE		0.015 / 300	TABLE : PALLET 1 : 0.003 PALLET 2 : 0.005 (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN XZ PLANE		0.015 / 300	TABLE : PALLET 1 : 0.004 PALLET 2 : 0.006 (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.001

SIGN & DATE	BSR	IG	RP	07-02-2012	4 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

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Sign: Date: 21/3/12

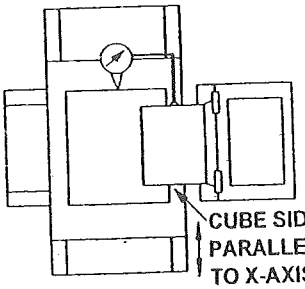
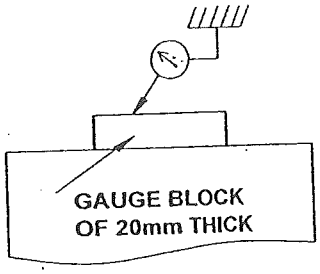
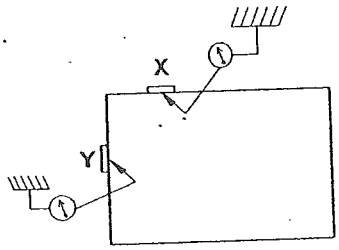
ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G13	SQUARENESS OF Y - AXIS MOVEMENT WITH RESPECT TO X - AXIS MOVEMENT	MEASURE READING ALONG 'Y' 	0.015 / 300	0.008
G14	VARIATION BETWEEN THE PALLET HEIGHT (FOR APC MACHINE ONLY)		0.030	0.010
G15	PALLET REPEATABILITY IN X & Y AXES WHEN LOADED ON THE MACHINE FROM THE PALLET STATION FOR SAME PALLET (FOR APC MACHINE ONLY)		0.020 IN X & Y AXES DIRECTION	PALLET 1 : 0.001 PALLET 2 : 0.002 (FOR APC MACHINE ONLY)

SIGN & DATE	BSR	IG	RP	07-02-2012	5 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

1) HEAVY CUT REPORT :

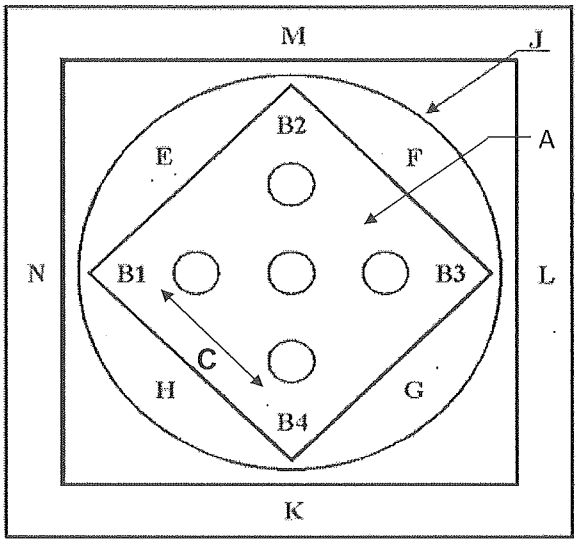
MODEL	: WINNER	BASE SPEED OF MOTOR	:1500 rpm	MATERIAL	: C.I.(Gr 25)
M/C NO.	: M26-848	SPINDLE SPEED RANGE	:60-6000 rpm	CUTTER	: Ø80 Sq.Sh FMC
TYPE OF MOTOR	: β i3/10000	MOTOR SPEED RANGE	:90-9000 rpm	WIDTH OF CUT	: 55 mm
PULLEY RATIO	: 1:1.5	MOTOR CAPACITY	:5.5/3.7 kW	C.S. (V)	: 250 mtr/min.

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as per CRT	Spl.load % as per CRT	Remarks
1000	0.15	1.05	1050	1.0	58	28	57	SMOOTH CUTTING
1000	0.15	1.05	1050	2.9	165	50	120	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NAS TEST REPORT :

Sl. No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings in mm
1	Flatness of Surface	A	0.015	0.005
2	Parallelism of Surfaces	N-L	0.020	0.005
		M-K	0.020	0.002
		H-F	0.030	0.002
		G-E	0.030	0.005
3	Squareness of side faces	K-L	0.020	0.004
		L-M	0.020	0.002
		M-N	0.020	0.004
		N-K	0.020	0.003
		E-F	0.030	0.005
		F-G	0.030	0.002
		G-H	0.030	0.004
		H-E	0.030	0.004
4	Circularity of Machining	J	0.040	0.013
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	-0.004
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	-0.002
	Diagonal pitch	C(70.71)	0.035	-0.002
6	Bore size variation		0.020	
	Hole No.	Max.	Min.	Ovality
	B1=	39.183	39.178	0.005
	B2=	39.185	39.182	0.003
	B3=	39.190	39.185	0.005
	B4=	39.210	39.208	0.002
7	Straightness	K	0.010	0.003
		L	0.010	0.002
		M	0.010	0.001
		N	0.010	0.003
		E	0.015	0.003
		F	0.015	0.001
		G	0.015	0.004
		H	0.015	0.002
8	Step difference in axis change over		0.010	0.002



SINGLE AXIS CONTROL K=L=M=N = 240 mm

DOUBLE AXIS CONTROL E=F=G=H = 160 mm

CIRCLE " J " = Ø 232.5mm

Heavy cut Done By	NAS TEST DONE BY	Department incharge	QC / CMM
manju.m	DEEPAK SARAN		
Date: 18/12/2012	DATE:08/10/2012	Date:18/12/2012	Date: 18/12/12

ACE MANUFACTURING SYSTEMS LTD

HYDRAULIC ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : M26 - 848 SHRI Ganesha 2 STN APC Power Pack

I. Power Pack Details			II. Pump Details	
1	Make :	Pascal / Hyd Tech / Rexorth./ Yuken /Jai Chamundi	Pump Make :	Rexroth / yuken /Corotech / vickers
2	Sl No. :	JCT-046-2012	Pump Sl. No. :	MFG NO 1230
3	Tank capacity :	32/ 40 / 63 / 70 Liters	Motor Make :	Rotomotive / Siemens
4	Cooler make :	Pascal System Technik Pvt Ltd	Motor Sl. No :	N° M07120978
5	Cooler fan make :	EBM NADI	—	—
6	No.of stations :	1 / 2 / 3 / 4 / 5 / 6	—	—

III	CHECK POINTS	Remark
1	Ensure Power pack is assembled as per circuit diagram & same is attached to inspection report.	✓
2	Ensure Accessories like Minnimess coupling, Guages are assembled.	✓
3	Ensure instruction name plates, power pack details, name plate are fitted	✓
4	All ports are tied with identification plate.	✓
5	Ensure there is no visual damage to the elements.	✓
6	Ensure oil level (at level indicator) in the hyd tank is at satisfactory	✓
7	Ensure tightness of all screws & marking neatly with yellow marker.	✓

IV	RUNNING CHECK POINTS	Remark
1	Check Motor direction is as per arrow marking.	✓
2	Ensure direction of rotation of cooler fan is as per arrow marking	✓
3	Ensure hydraulic pressure switch set as per circuit diagram.	✓
4	Ensure Accumulator oil & gas pressure as per circuit diagram (if applicable).	NA
5	Ensure no oil leakage from hydraulic system including at the oil level indicator & pipe joints	✓
6	Ensure continuous running of power pack for 4 hrs with solenoid actuation once per minute	✓
7	Power pack oil temp rise is within 30°C above ambient temp or 65 °C maximum.	✓
8	Check for any abnormal / unhealthy noise	✓

REMARK IF ANY:

25/09/12 Date:	Checked by	Approved by	

ACE MANUFACTURING SYSTEMS LTD

SPINDLE BALANCING REPORT

AMS — In Pursuit of Excellence

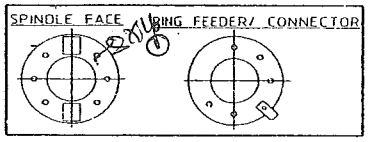
Machine Number : M26 - 848

Spindle Serial Number : 17142

Spindle Type : Direct

Spindle Speed (Max.) : 8000 rpm

Note : Refer TM-QA-09 for measurement procedure.



Location of balancing screw(s)

Sl No.	Speed in RPM	Velocity of vibration (mm/sec) before Balancing Along X-axis	After Balancing along X-axis (8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.)	After Balancing along Y-axis (8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.)
1	500			
2	1000	0.029	0.015	0.005
3	1500	0.029	0.007	
4	2000	0.030	0.009	0.016
5	2500	0.080	0.011	
6	3000	0.080	0.007	0.012
7	3500	0.271	0.058	
8	4000	0.600	0.186	0.050
9	4500	1.36 - 323°	0.366	
10	5000	1.54 - 298°	0.512	0.098
11	5500	1.05 - 251°	0.346	
12	6000	1.46 - 251°	0.443	0.191
13	6500			
14	7000			
15	7500			
16	8000			
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle maximum vibration mm / sec :

Correction in gms :

REMARK IF ANY: Added the mas to top position M6 x 10 - ①

Date: 08/10/12

Checked By

APPROVED BY

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

Machine No	:-	M26-848.	Date :- 2012-09-29 15:49
Machine Modle	:-	WINNER.	Cond by:- QAS/RKU.

AXIS CALIBRATION PITCH ERROR REPORT

X				Y				Z			
SL NO 100 (X)	0			SLNO 200 (Y)	0			SL NO 300 (Z)	0		
101	1	141	0	201	0	241	0	301	0	341	0
102	0	142	0	202	0	242	0	302	1	342	0
103	-1	143	1	203	0	243	0	303	1	343	0
104	-1	144	1	204	-1	244	0	304	0	344	0
105	0	145	0	205	-1	245	0	305	0	345	0
106	0	146	0	206	0	246	0	306	0	346	0
107	1	147	0	207	1	247	0	307	0	347	0
108	0	148	0	208	0	248	0	308	0	348	0
109	-1	149	0	209	0	249	0	309	0	349	0
110	0	150	0	210	-1	250	0	310	1	350	0
111	0	151	0	211	-1	251	0	311	0	351	0
112	2	152	1	212	0	252	0	312	0	352	0
113	0	153	0	213	-1	253	0	313	0	353	0
114	-1	154	0	214	0	254	0	314	0	354	0
115	-1	155	0	215	1	255	0	315	0	355	0
116	0	156	0	216	1	256	0	316	0	356	0
117	1	157	0	217	-1	257	0	317	1	357	0
118	1	158	0	218	0	258	0	318	1	358	0
119	1	159	0	219	0	259	0	319	1	359	0
120	2	160	0	220	-1	260	0	320	1	360	0
121	1	161	0	221	0	261	0	321	1	361	0
122	0	162	0	222	0	262	0	322	1	362	0
123	1	163	0	223	1	263	0	323	0	363	0
124	0	164	0	224	2	264	0	324	1	364	0
125	0	165	0	225	0	265	0	325	1	365	0
126	1	166	0	226	0	266	0	326	1	366	0
127	0	167	0	227	0	267	0	327	2	367	0
128	1	168	0	228	0	268	0	328	1	368	0
129	0	169	0	229	0	269	0	329	0	369	0
130	0	170	0	230	0	270	0	330	0	370	0
131	1	171	0	231	0	271	0	331	0	371	0
132	0	172	0	232	1	272	0	332	1	372	0
133	-1	173	0	233	0	273	0	333	1	373	0
134	-1	174	0	234	0	274	0	334	0	374	0
135	0	175	0	235	0	275	0	335	0	375	0
136	1	176	0	236	-2	276	0	336	0	376	0
137	0	177	0	237	-1	277	0	337	0	377	0
138	0	178	0	238	0	278	0	338	0	378	0
139	0	179	0	239	-1	279	0	339	0	379	0
140	0	180	0	240	1	280	0	340	0	380	0

SL NO	PARARMETER NO	DISCRIPTION	X	Y	Z
1	3620	P COMP REF NO	152	240	335
2	3621	P COMP MIN NO	100	200	300
3	3622	P COMP MAX NO	157	245	340
4	3623	PCOMP MULTIPLICATION	10	10	10
5	3624	P COMP INTERVAL	10	10	10

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

M/c NO :- M26-848.	COND BY:- QAS/RKU.
Machine Modle :- WINNER.	DATE:- 30-Sep-12

AXIS CALIBRATION BALL BAR PARAMETERS

SERVO PARAMETERS RELATED TO CIRCULARITY FOR FANUC SYSTEM

SL No.	PARAMETER No.	TYPICAL VALUE	DESCRIPTION	X	Y	Z	REMARKS
1	1851	##	BACKLASH IN FEED	95	205	130	
2	1852	##	BACKLASH IN RAPID	55	80	75	
3	1825	8000	SERVO LOOP GAIN	5000	5000	5000	
4	1826	500	IN POSITION WIDTH	500	500	500	
5	2021	256	LOAD INERTIA	251	215	230	******
6	2048	600 (###)	BACK ACCEL COMPENSATION VALUE	30	75	50	
7	2094	600 (###)	BACK ACCEL COMPENSATION VALUE	30	75	50	
8	2066	10 TO -20	250 MICRO SEC ACC FEEDBACK	-10	-10	-10	******
9	2068	70 TO700	FEED FWD CO- EFFICIENT	100	100	100	
10	2069	50 TO 200	VELOCITY FEED FWD CO- EFFICIENT FOR RAPID	100	100	100	
11	2071	50 TO 200	TIME DURING WHICH BACK ACCELERATION IS EFFECTIVE	100	100	100	
12	2145	100	VELOCITY FEED FWD CO- EFFICIENT FOR CUTTING	100	100	100	SHOULD NOT BE CHANGED
13	2144	9900	LOOK AHEAD FFW COEFFICIENT FOR CUTTING	9900	9850	9850	
14	2092	9200	LOOK AHEAD FFW COEFFICIENT FOR RAPID	7000	7000	7000	
15	2082	100	STAGE 2 START / END PARAMETER	100	100	100	
16	2039	100	STAGE 2 ACCELERATION AMOUNT	300	300	300	
17	1622	NA	CUT TIME CONSTANT	60	60	60	Same value for 3 axis

NOTE:- ***** DON'T CHANGE THE VALUE

VARIES FOR ALL MACHINES

Ballbar diagnostics table
M26-848-XY-B

Operator: Shankar.
Date: 2012-Sep-30 09:52:07

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸-0.7	◂-0.1 µm	0.7 µm	(3%)	(12)
Backlash Y	▴2.2	▾1.3 µm	2.2 µm	(8%)	(6)
Reversal spikes X	▸-2.8	◂-2.1 µm	2.8 µm	(10%)	(4)
Reversal spikes Y	▴-7.1	▾-6.5 µm	7.1 µm	(25%)	(1)
Lateral play X	▸-1.9	◂0.1 µm	1.2 µm	(4%)	(8)
Lateral play Y	▴-2.3	▾2.3 µm	1.1 µm	(4%)	(9)
Cyclic error X	↑1.2	↓1.1 µm	1.1 µm	(4%)	(10)
Cyclic error Y	↑1.2	↓1.5 µm	1.4 µm	(5%)	(7)
Servo mismatch	0.05 ms		0.9 µm	(3%)	(11)
Squareness	24.2 µm/m		3.6 µm	(13%)	(2)
Straightness X	-4.8 µm		2.4 µm	(8%)	(5)
Straightness Y	0.0 µm		0.0 µm	(0%)	(13)
Scaling mismatch	-7.2 µm		3.6 µm	(13%)	(3)
Cyclic pitch X	50.8000 mm				
Cyclic pitch Y	40.0000 mm				
Scaling error X	4.0 ppm				
Scaling error Y	27.9 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset X	0.2 µm				
Centre offset Y	0.1 µm				
Positional tolerance	25.7 µm				
Best fit radius	150.0024 mm				
Circularity	13.2 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M26-848-XY-B

Operator: Shankar.
Date: 2012-Sep-30 09:52:07

RENISHAW

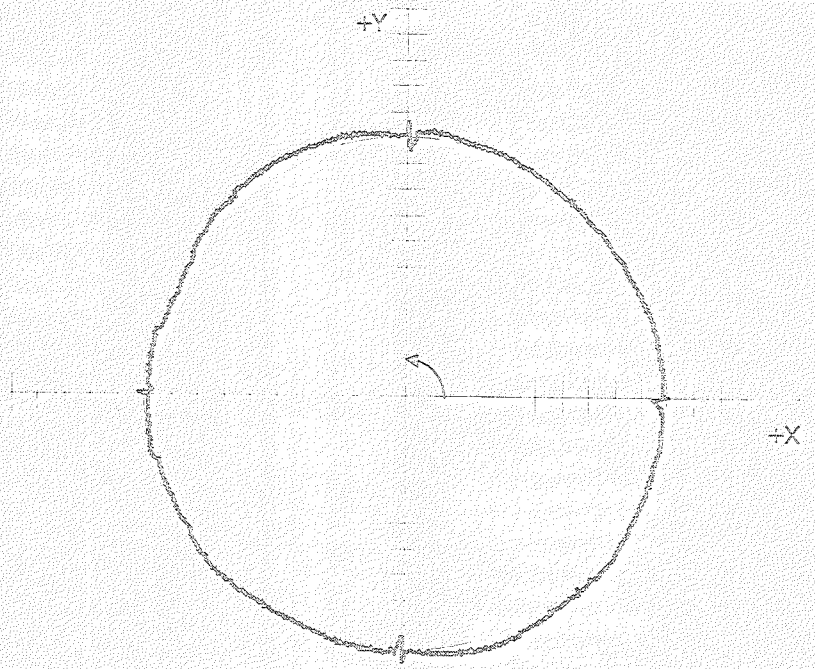
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	12.1µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M26-848-XY-B

Operator: Shankar.
Date: 2012-Sep-30 09:52:07

RENISHAW

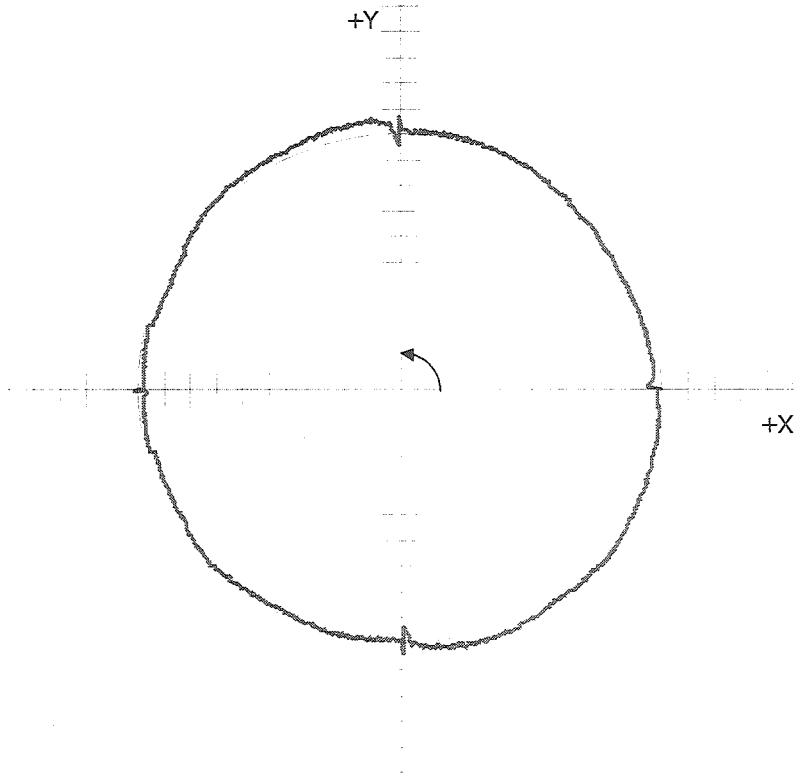
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	13.0µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0 µm/div

Ballbar diagnostics table
M26-848-XZ-B

Operator: Shankar.
Date: 2012-Sep-30 10:22:13

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▵0.4	▿1.8 µm	1.8 µm	(8%)	(6)
Backlash X	▹1.0	◃0.5 µm	1.0 µm	(4%)	(11)
Reversal spikes Z	▵-0.9	▿-1.2 µm	1.2 µm	(5%)	(10)
Reversal spikes X	▹-2.8	◃-2.0 µm	2.8 µm	(12%)	(2)
Lateral play Z	▵1.7	▿2.9 µm	2.4 µm	(10%)	(3)
Lateral play X	▹-2.9	◃-0.5 µm	2.0 µm	(8%)	(5)
Cyclic error Z	↑1.5	↓1.6 µm	1.4 µm	(6%)	(8)
Cyclic error X	↑1.7	↓1.8 µm	1.7 µm	(7%)	(7)
Servo mismatch	-0.06 ms		0.9 µm	(4%)	(12)
Squareness	15.8 µm/m		2.4 µm	(10%)	(4)
Straightness Z	2.6 µm		1.3 µm	(5%)	(9)
Straightness X	8.3 µm		4.1 µm	(17%)	(1)
Scaling mismatch	1.6 µm		0.8 µm	(3%)	(13)
Cyclic pitch Z	50.0000 mm				
Cyclic pitch X	45.0000 mm				
Scaling error Z	16.0 ppm				
Scaling error X	10.8 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset Z	-6.8 µm				
Centre offset X	-0.2 µm				
Positional tolerance	25.2 µm				
Best fit radius	150.0020 mm				
Circularity	12.4 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M26-848-XZ-B

Operator: Shankar.
Date: 2012-Sep-30 10:22:13

RENISHAW

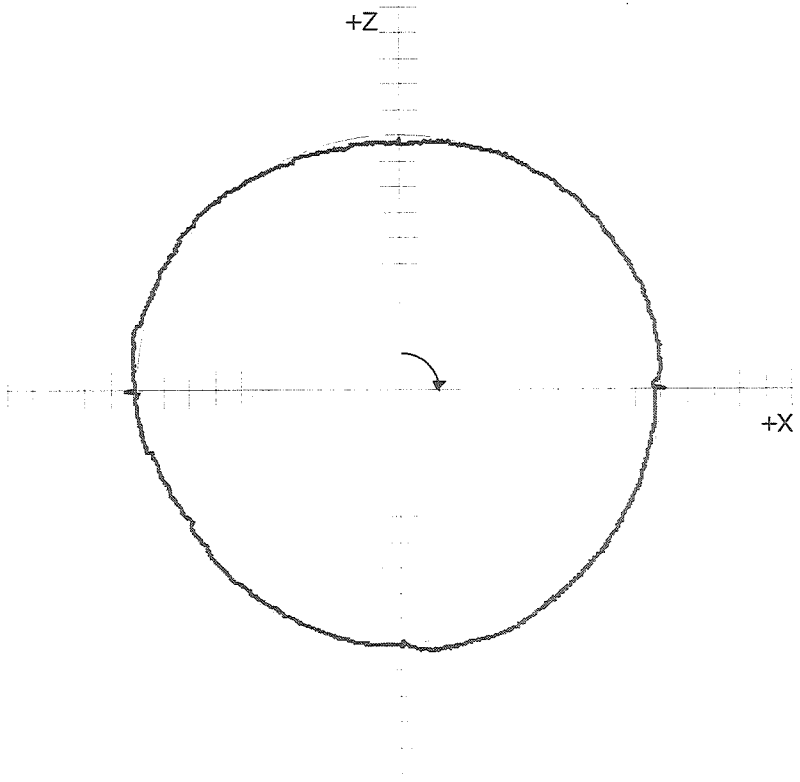
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	9.9µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

10.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M26-848-XZ-B

Operator: Shankar.
Date: 2012-Sep-30 10:22:13

RENISHAW

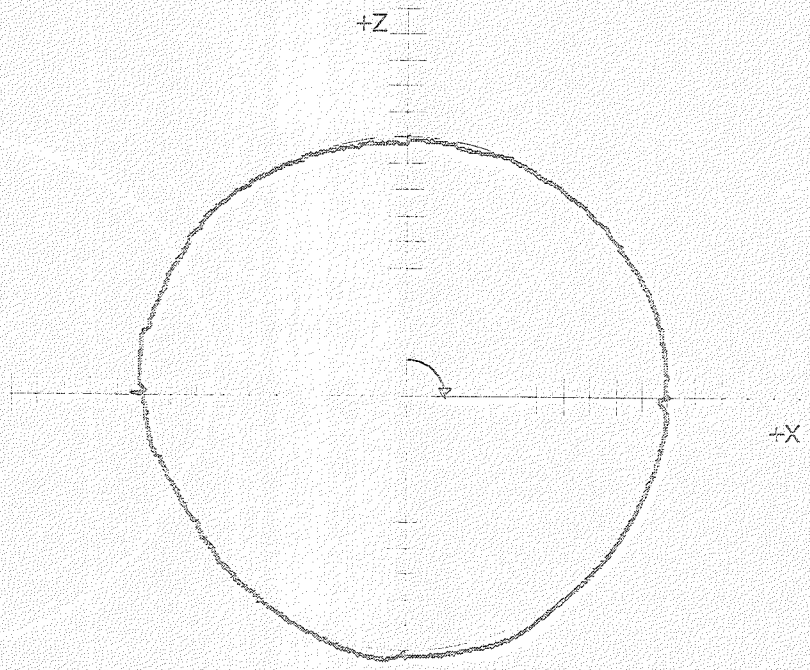
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	10.7µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

10.0 µm/div

Ballbar diagnostics table
M26-848-YZ-A

Operator: Shankar.
Date: 2012-Sep-30 10:37:57

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude			Independent circularity	Ranking
Backlash Y	▸4.3	◂ 2.7 µm	4.3 µm	(13%)	(3)
Backlash Z	▴2.2	▾4.3 µm	4.3 µm	(13%)	(2)
Reversal spikes Y	▸-6.7	◂ -5.0 µm	6.7 µm	(21%)	(1)
Reversal spikes Z	▴-0.6	▾-0.9 µm	0.9 µm	(3%)	(9)
Lateral play Y	▸-1.6	◂ 0.4 µm	1.0 µm	(3%)	(8)
Lateral play Z	▴0.2	▾0.9 µm	0.7 µm	(2%)	(12)
Cyclic error Y	↑1.5	↓1.7 µm	1.6 µm	(5%)	(7)
Cyclic error Z	↑0.9	↓0.8 µm	0.8 µm	(3%)	(10)
Servo mismatch	0.03 ms		0.6 µm	(2%)	(13)
Squareness	-24.7 µm/m		3.7 µm	(12%)	(5)
Straightness Y	7.5 µm		3.8 µm	(12%)	(4)
Straightness Z	1.5 µm		0.7 µm	(2%)	(11)
Scaling mismatch	6.3 µm		3.1 µm	(10%)	(6)
Cyclic pitch Y	44.4500 mm				
Cyclic pitch Z	25.0000 mm				
Scaling error Y	37.5 ppm				
Scaling error Z	16.6 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset Y	0.1 µm				
Centre offset Z	0.0 µm				
Positional tolerance	35.2 µm				
Best fit radius	150.0041 mm				
Circularity	13.5 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation

M26-848-YZ-A

Operator: Shankar.
Date: 2012-Sep-30 10:37:57

RENISHAW

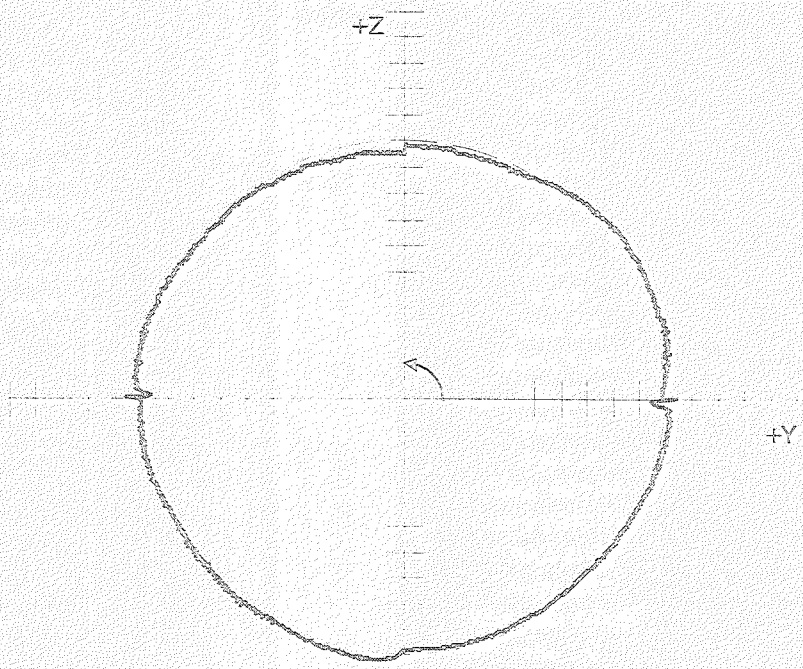
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	12.0µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M26-848-YZ-A

Operator: Shankar.
Date: 2012-Sep-30 10:37:57

RENISHAW

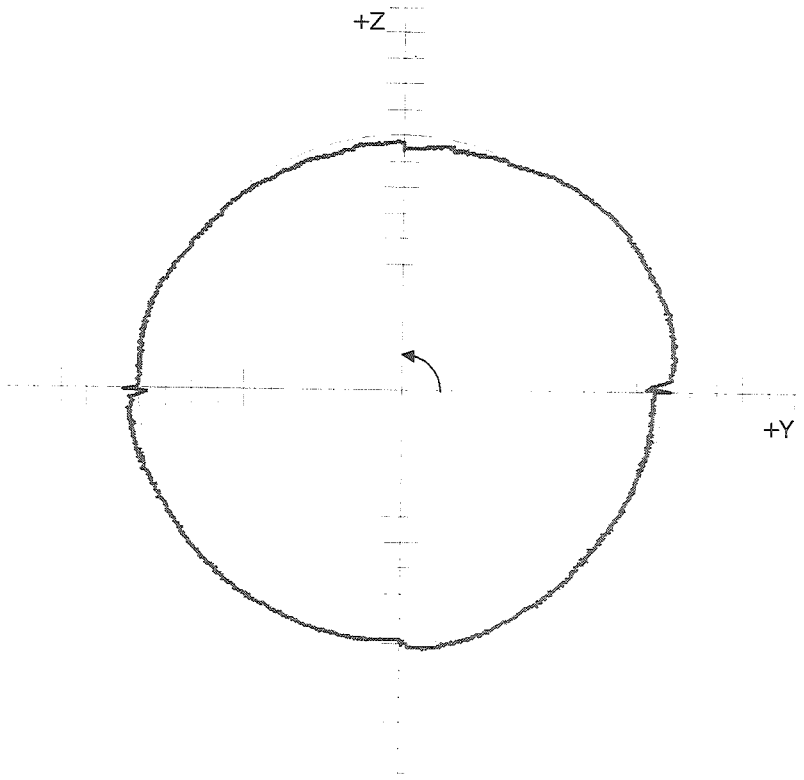
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	13.1µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0 µm/div

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M26-848-X.RTL

Machine:WINNER.
Date:2012-09-29 16:21
Axis:X.

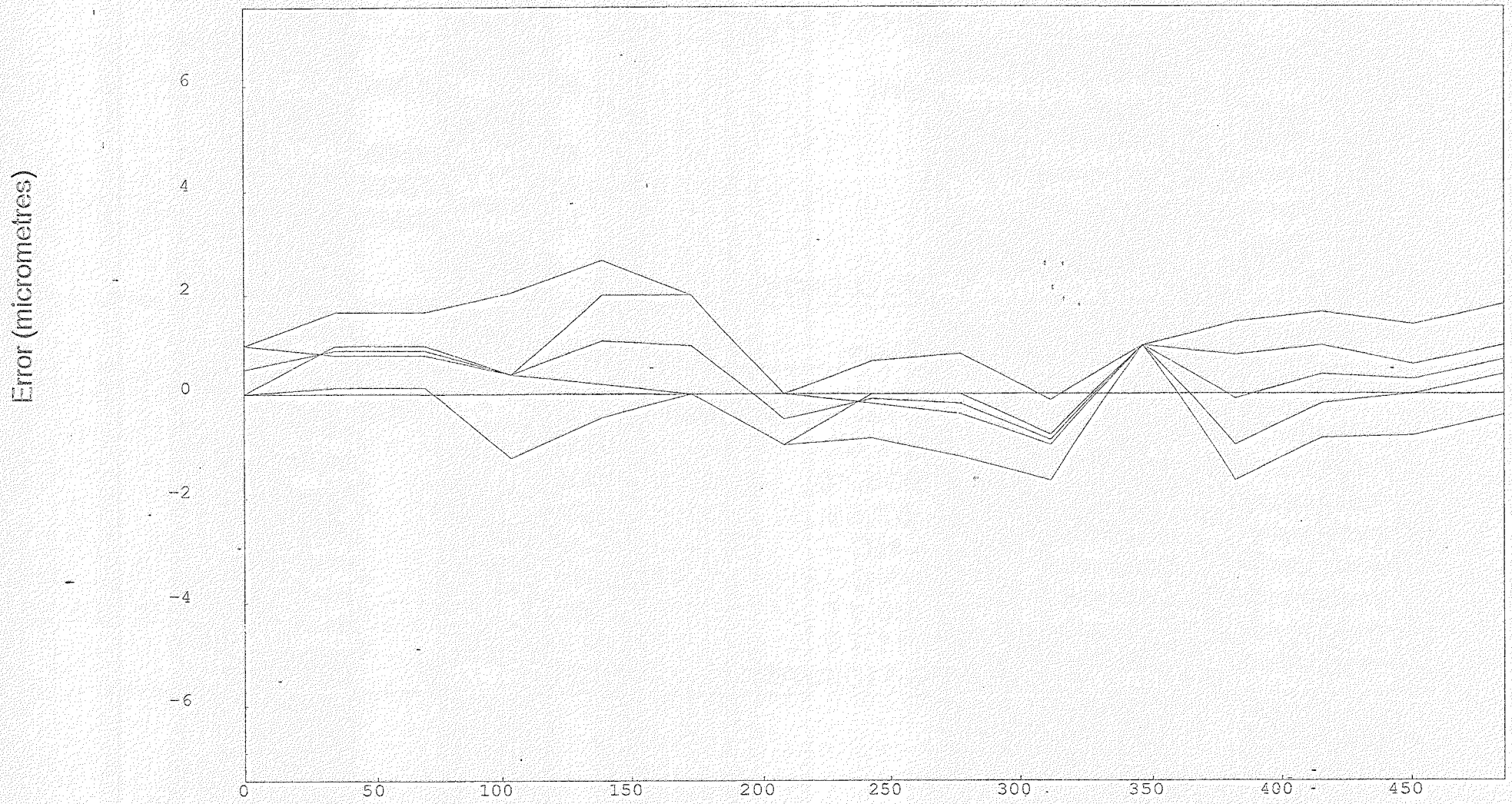
Serial No:M26-848.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	2012-09-29 16:21	2012-09-29 16:30	
Air temperature :	27.02	26.97	deg C
Air pressure :	905.40	905.40	mbar
Rel. humidity :	57.58	57.76	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	26.17	26.18	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639767	0.31639763	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:WINNER.	Axis:X.	U max : 2.000
Serial No:M26-848.	Location:AMS.	P : 4.342
Date:2012-09-29 16:21	U mean : 0.747	Ps max : 3.286
By:QAS/RKU.	Ps mean : 1.174	Pa : 2.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:WINNER. Serial No:M26-848.
Date:2012-09-29 16:21 By:QAS/RKU.
Axis:X. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M26-848-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 2 (+) (34.6670 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 2 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

Target 3 (+) (69.3340 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 3 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

Target 4 (+) (104.0010 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 1.0

Target 5 (+) (138.6680 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 6 (+) (173.3350 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 7 (+) (208.0020 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (+) (242.6690 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (277.3360 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) -1.0 2) 0.0 3) -1.0 4) 0.0
5) 0.0

Target 10 (+) (312.0030 millimetres)
1) -1.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (346.6700 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 11 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 12 (+) (381.3370 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 12 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (416.0040 millimetres)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (450.6710 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (485.3380 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M26-848-Y-1.RTL

Machine:WINNER.
Date:2012-12-18 12:23
Axis:Y.

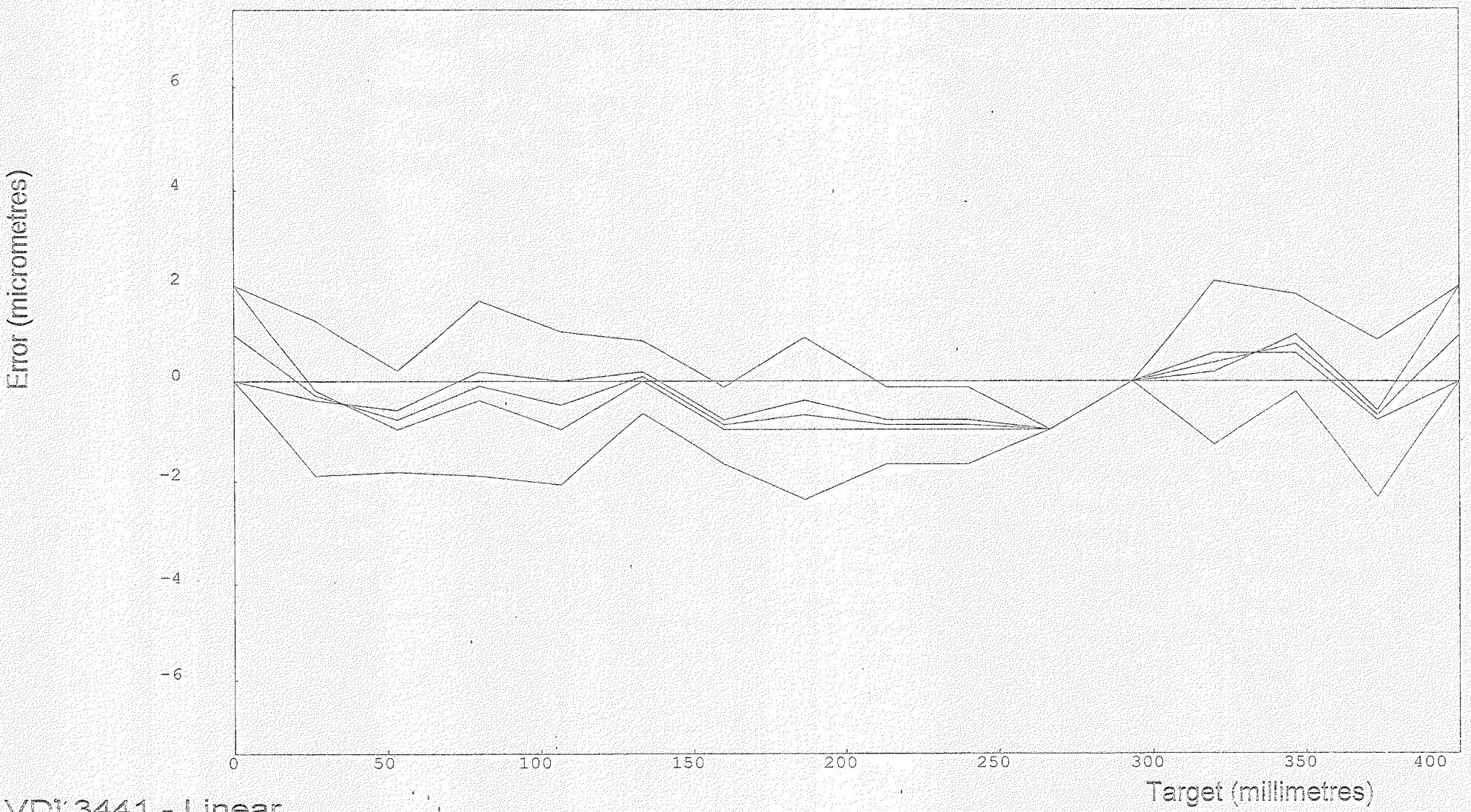
Serial No:M26-848.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	2012-12-18 12:24	2012-12-18 12:33	
Air temperature :	23.87	24.14	deg C
Air pressure :	912.40	912.20	mbar
Rel. humidity :	59.15	57.84	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	21.59	21.67	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641320	0.31641299	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:WINNER.	Axis:Y.	U max : 2.000
Serial No:M26-848.	Location:AMS.	P : 4.434
Date:2012-12-18 12:23	U mean : 0.537	Ps max : 2.985
By:QAS/RKU.	Ps mean : 1.587	Pa : 2.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:WINNER. Serial No:M26-848.
Date:2012-12-18 12:23 By:QAS/RKU.
Axis:Y. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M26-848-Y-1.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 2 (+) (26.6670 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 3 (+) (53.3340 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 3 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 4 (+) (80.0010 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) -0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0

Target 5 (+) (106.6680 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) -1.0
5) 0.0
Target 5 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (133.3350 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (160.0020 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (186.6690 millimetres)
1) 1.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (213.3360 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 10 (+) (240.0030 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (266.6700 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 11 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 12 (+) (293.3370 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 13 (+) (320.0040 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (346.6710 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (373.3380 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 15 (-)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0

Target 16 (+) (400.0050 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M26-848-Z.RTL

Machine:WINNER.
Date:2012-09-29 18:41
Axis:Z.

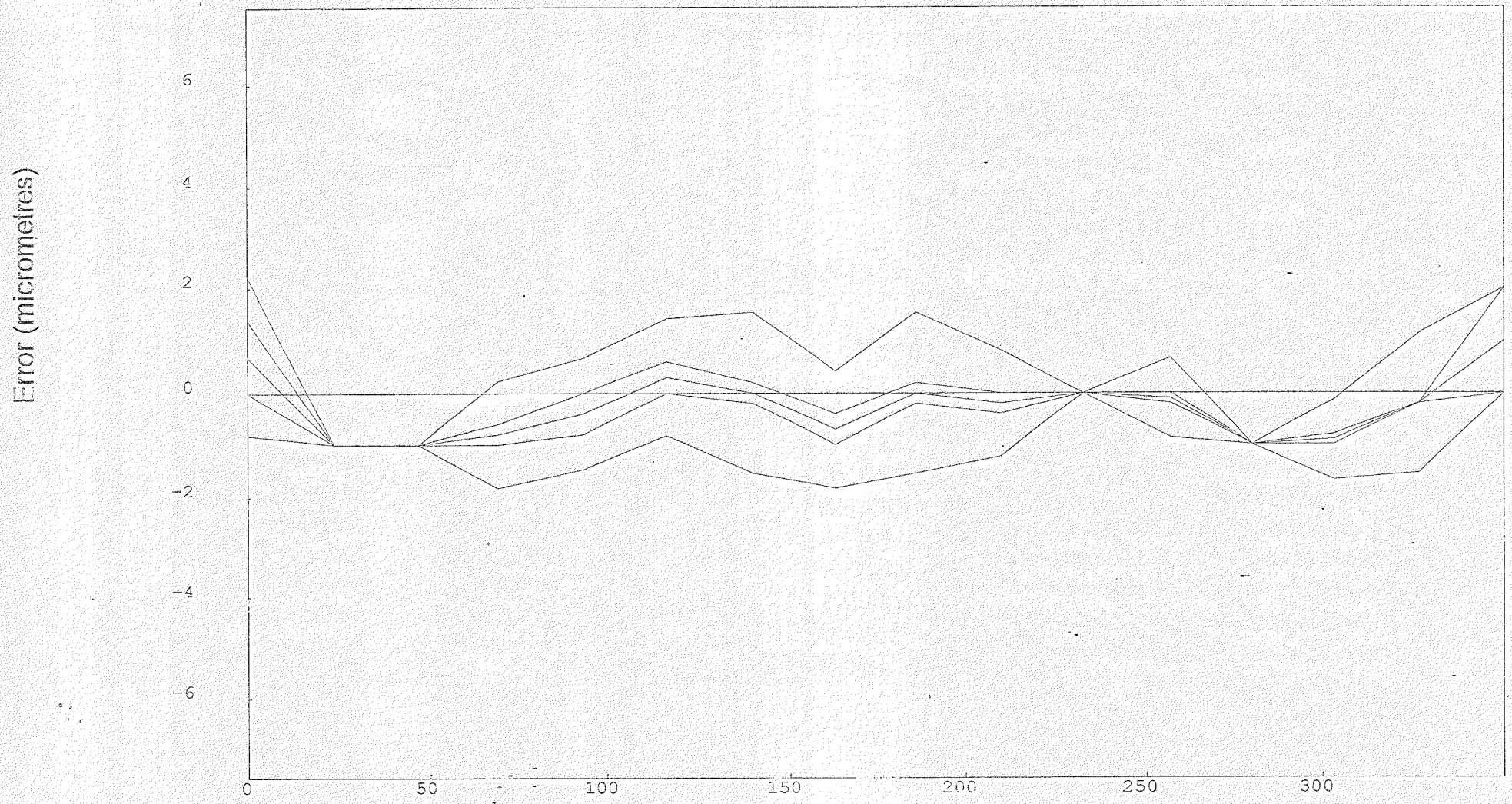
Serial No:M26-848.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	2012-09-29 18:42	2012-09-29 18:51	
Air temperature :	26.18	26.16	deg C
Air pressure :	906.80	907.00	mbar
Rel. humidity :	60.85	62.15	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	26.21	26.20	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639720	0.31639724	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:WINNER.	Axis:Z.	U max : 2.000
Serial No:M26-848.	Location:AMS.	P : 4.043
Date:2012-09-29 18:41	U mean : 0.462	Ps max : 2.683
By:QAS/RKU.	Ps mean : 1.268	Pa : 2.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:WINNER. Serial No:M26-848.
Date:2012-09-29 18:41 By:QAS/RKU.
Axis:Z. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M26-848-Z.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 2.0 2) 2.0 3) 1.0 4) 1.0
5) 1.0

Target 2 (+) (23.3330 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 3 (+) (46.6660 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 3 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 4 (+) (69.9990 millimetres)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0
Target 4 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (93.3320 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (116.6650 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (139.9980 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 8 (+) (163.3310 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0

Target 8 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (186.6640 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 10 (+) (209.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0

Target 11 (+) (233.3300 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 12 (+) (256.6630 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
-Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 13 (+) (279.9960 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 13 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 14 (+) (303.3290 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 14 (-)
1) -1.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 15 (+) (326.6620 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0
Target 15 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 16 (+) (349.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY - INSPECTION REPORT- BBA

AMS In Pursuit of Excellence

MACHINE NUMBER: M-26- 848

V. Axis Motor Assembly:

Motor mounting bracket Concentricity & Face out:

Sl No	Axis	Face out -Recommended 0.030 mm	Concentricity Recommended 0.060 mm
1	X- Axis	0·020	0·030
2	Y- Axis	0·020	0·020
3	Z- Axis	0·010	0·030

SI NO	Check points:	Remark
1	Ensure motor shaft and coupling bore are free from oil / grease before fitting of coupling	Ok
2	Ensure Axis Motor & Motor mounting bracket are fitted as per specification.	Ok
3	Ensure that slot on Steel Bellows, Clamping ring & Bush are in line.	Ok
4	Ensure tightness of coupling is set to torque value of 17 N-m (M6 screw), 35 N-m (M8 screw).	Ok

VI. Counter balance Assembly (If applicable)

SI NO	Check points:	Remark
1	Ensure Smooth movement of Z- Axis without fouling of dead weight. (MCV 500 / MCV 600 only)	— N/A —
2	Ensure the parallelism of counter balance cylinder piston is within 0.050 mm in both XZ & YZ planes with respect to Z Axis movement.	— N/A —
3	Ensure variation in the vertical axis motor current is within 20% during upward and downward movement of slide.	— N/A —
4	Ensure dowelling of counterbalance bracket.	— N/A —

REMARK IF ANY:

08/09/12	C.S. Mehta		12.04.10
Date:	PREPARED	CHECKED	APPROVED

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY – INSPECTION REPORT- BBA

AMS *In Pursuit of Excellence*

MACHINE NUMBER : M-26- 8H8

I. Lubrication metering cartridge specification (cc of oil / shot) as per DS-PDN-MEC-LUB-01 :

Sl. No.	Axis	Ball screw	Box guideway top / LM block	Box guideway side	Jib	Keeper plate
1	X-Axis	0.25 cc	0.16 cc	0.16 cc	0.1 cc	0.1 cc
2	Y-Axis	0.25 cc	0.16 cc	0.16 cc	0.1 cc	0.1 cc
3	Z-Axis	0.25 cc	0.16 cc	0.16 cc	0.1 cc	0.1 cc

Sl. No.	Check points:	Remark
1	Verify that metering cartridge are fixed as per DS-PDN-MEC-LUB-01.	
2	Ensure X,Y & Z guideways and ball screw are lubricating properly	Ok
3	Ensure lubrication pressure set to 15 Bar.	Ok
4	Ensure there is no leakage of oil from joints, cartridges & manifolds etc.	Ok
5	Ensure that the lubrication hoses do not rub or foul with the stationary / moving parts during axis movement.	Ok
6	Ensure that the lubrication hoses of appropriate length (as per DS-PDN-MEC-LUB-01) are used and properly clamped and routed.	Ok

II Telescopic cover Assembly:

	Sl No	Axis	Extension block Parallelism w.r.t. axis guideway		Extn. Block Length Variation -	DRAWING NUMBER
			Top - Recommended 0.020 mm	Side- Recommended 0.030 mm	Recommended 0.050 mm	(S10 -)
	1	X – Motor End	0.010	0.020	0.040	
	2	X – Brake End	0.010	0.020	0.040	
	3	Y – Motor End	0.020	0.020	– NA –	
	4	Y – Brake End	0.020	0.010	0.040	
	5	Z – Brake End	0.010	0.020	0.050	

III. Saddle cover alignment w.r.t. Y axis movement (If applicable) :

	Sl NO	Top - Recommended 0.050 mm	Side - Recommended 0.050 mm	
	1	– NA –	– NA –	

IV. Check Points :

Sl NO	Check points:	Remark
1	Check telescopic cover sliding smoothly	Ok
2	Ensure telescopic cover screws are tightened in box close condition	Ok

Date: 14/09/12	C.S. Medha PREPARED	CHECKED	04/10 APPROVED